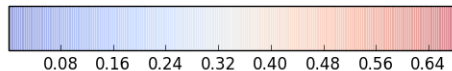
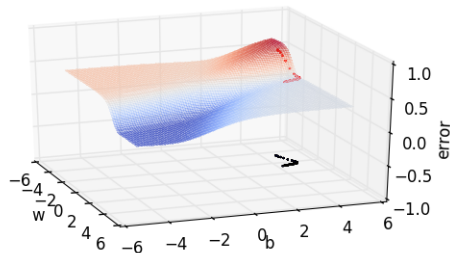
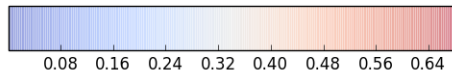
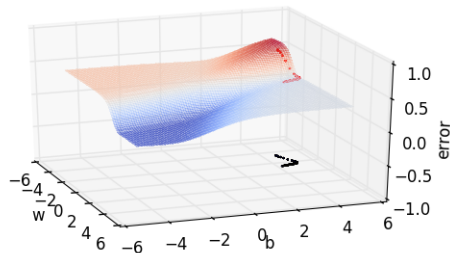


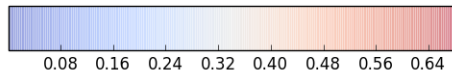
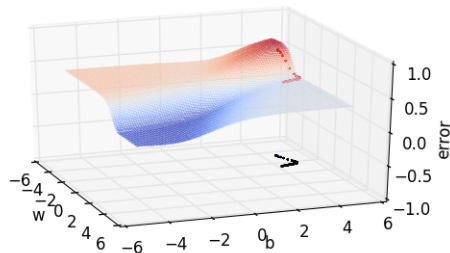
- Irrespective of where we start from once we hit a surface which has a gentle slope, the progress slows down



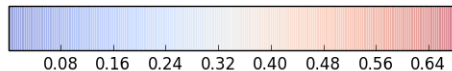
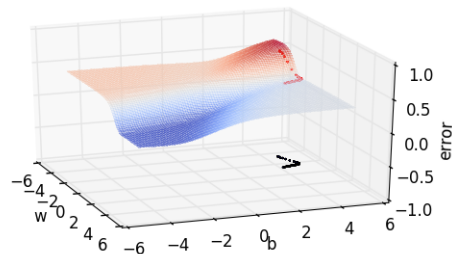
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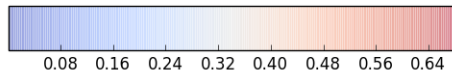
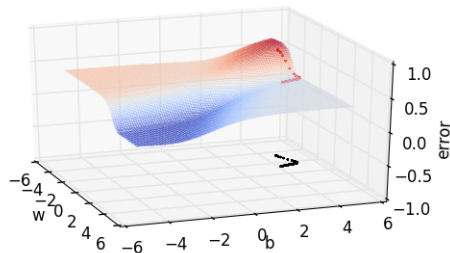
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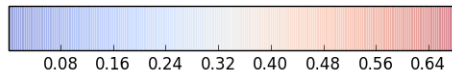
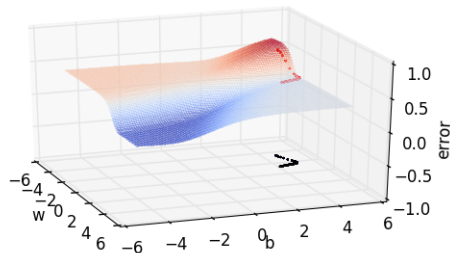
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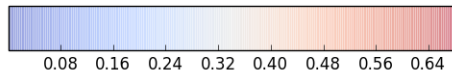
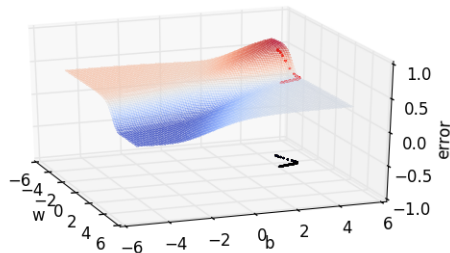
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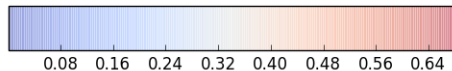
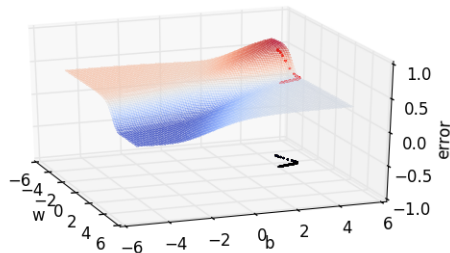
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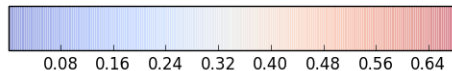
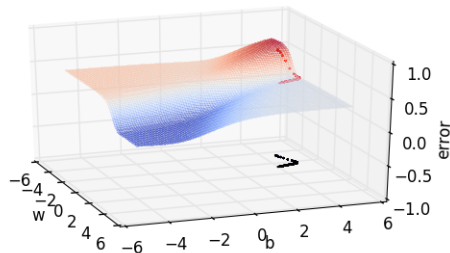
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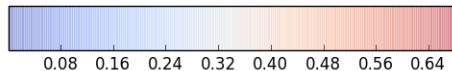
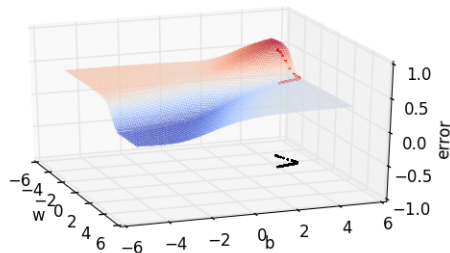
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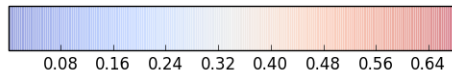
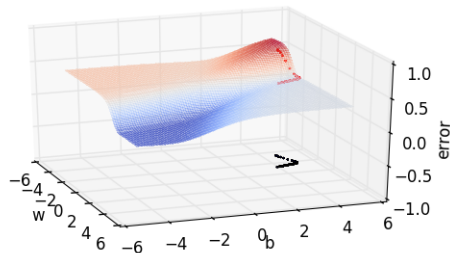
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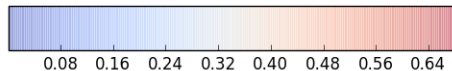
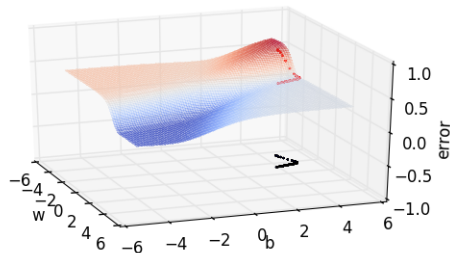
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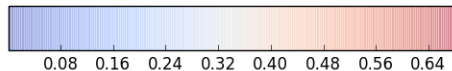
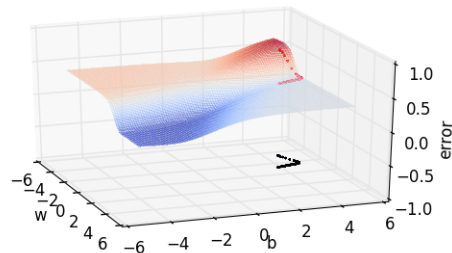
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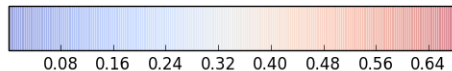
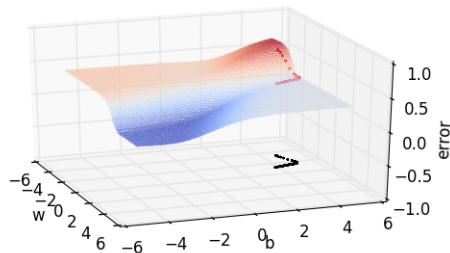
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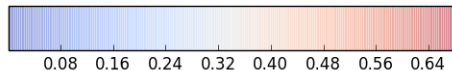
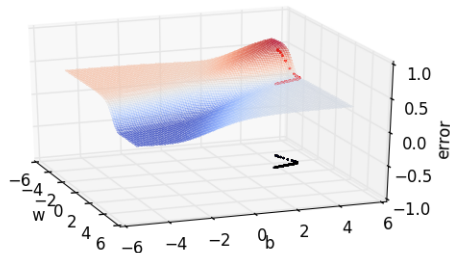
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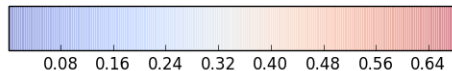
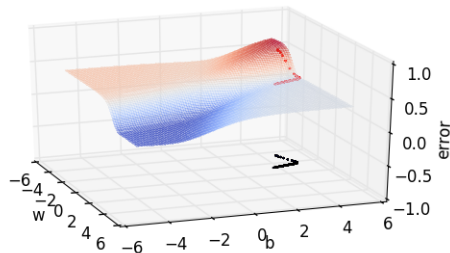
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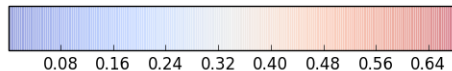
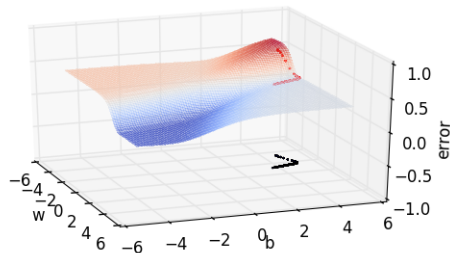
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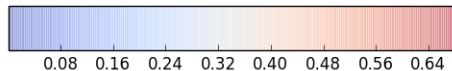
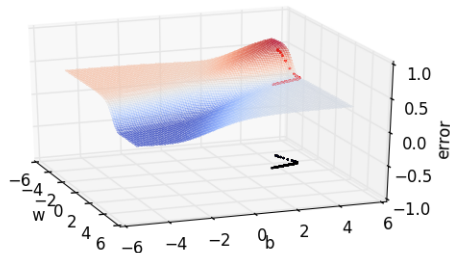
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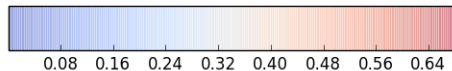
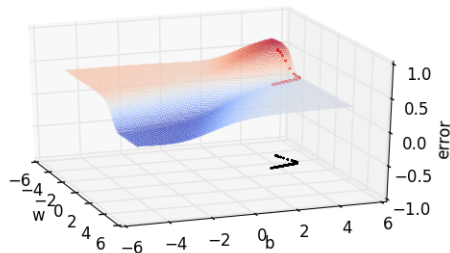
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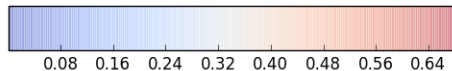
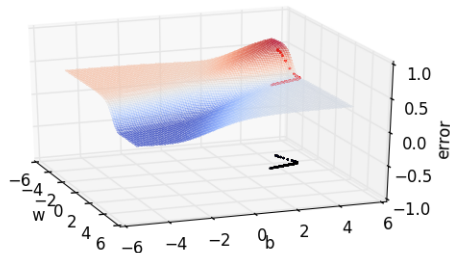
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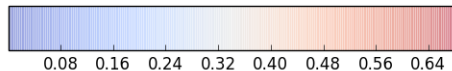
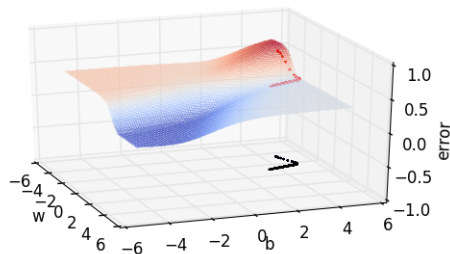
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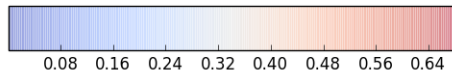
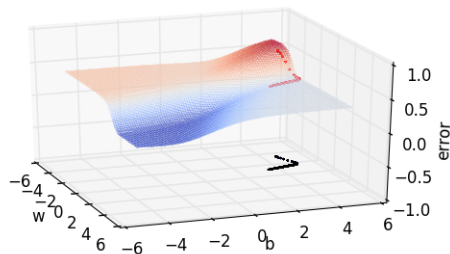
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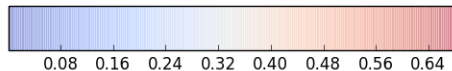
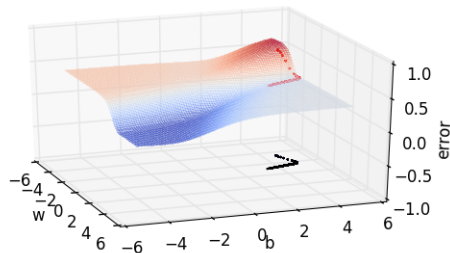
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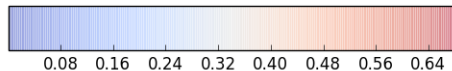
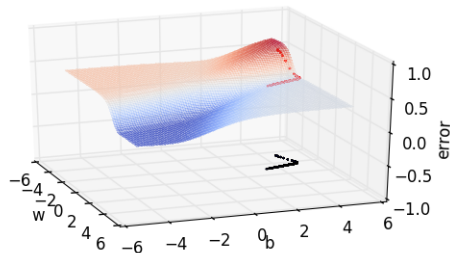
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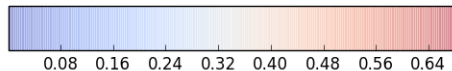
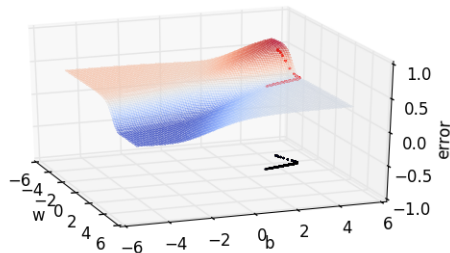
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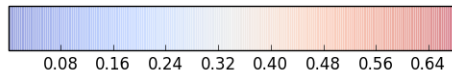
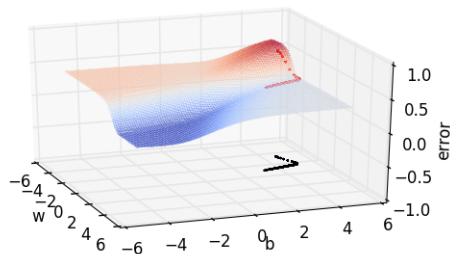
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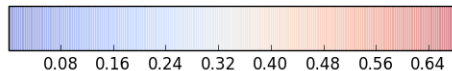
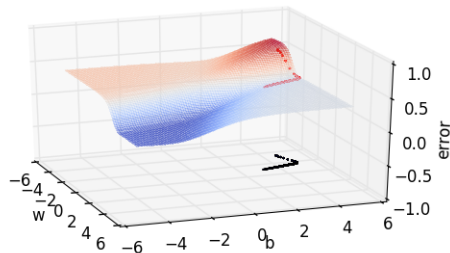
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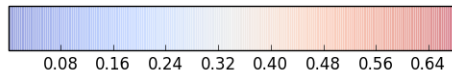
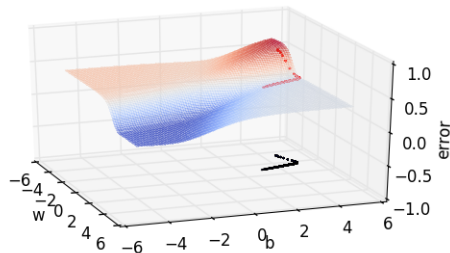
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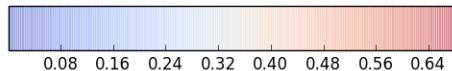
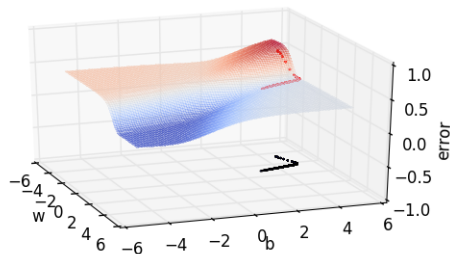
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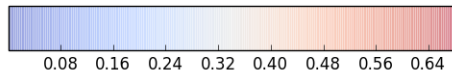
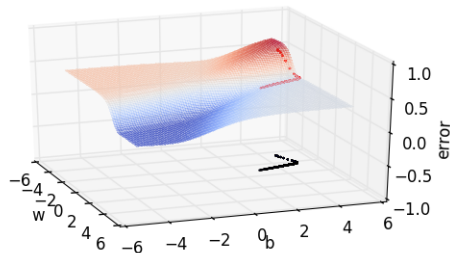
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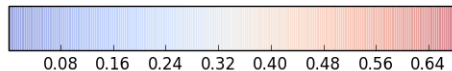
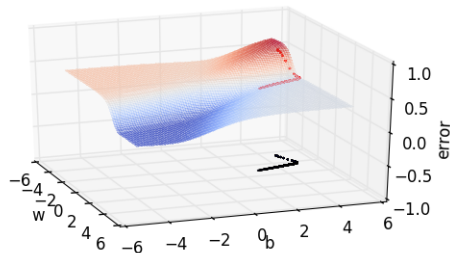
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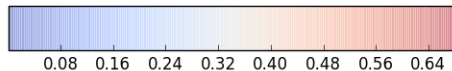
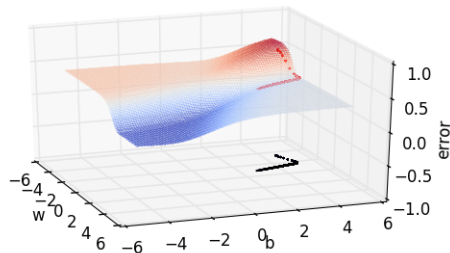
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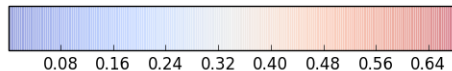
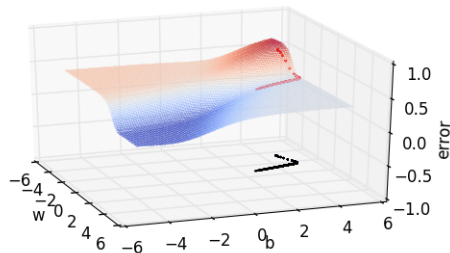
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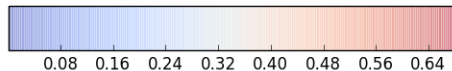
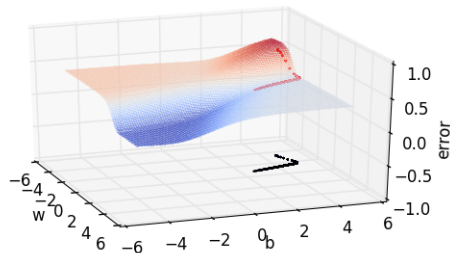
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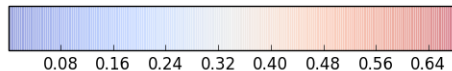
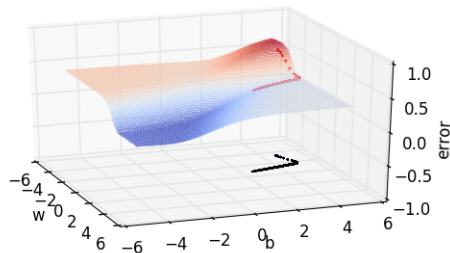
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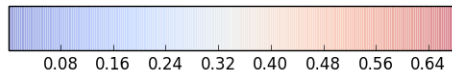
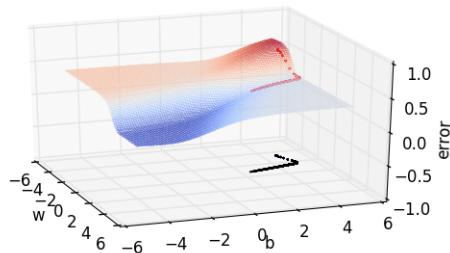
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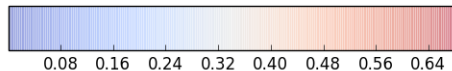
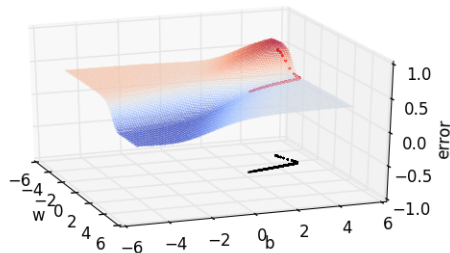
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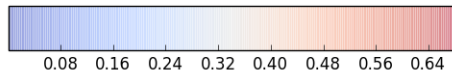
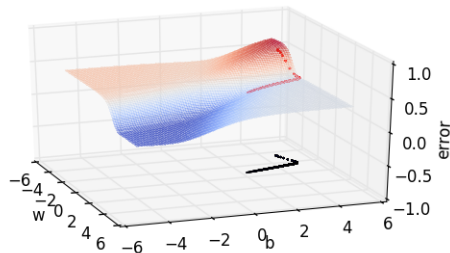
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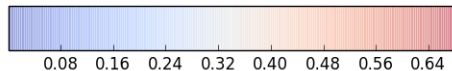
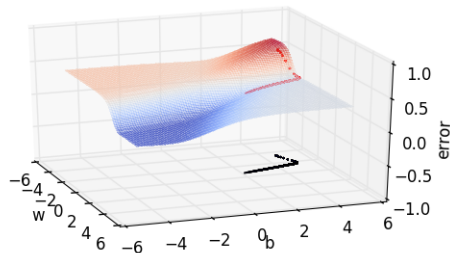
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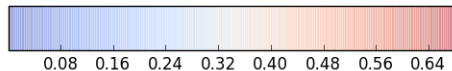
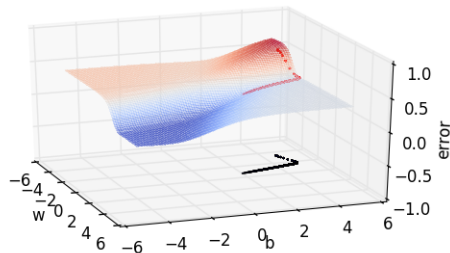
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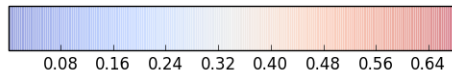
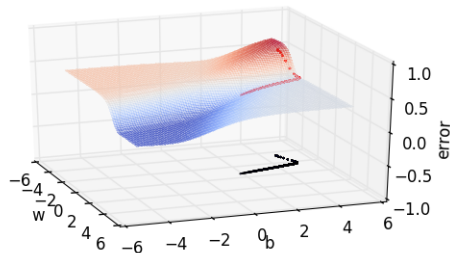
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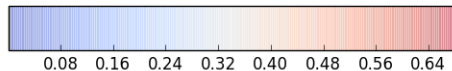
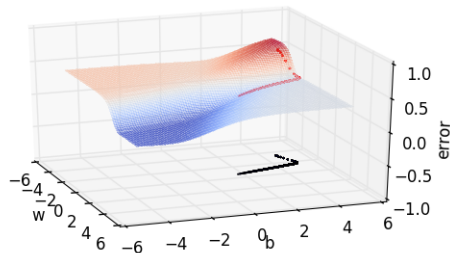
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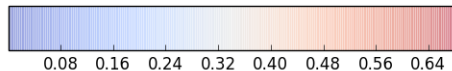
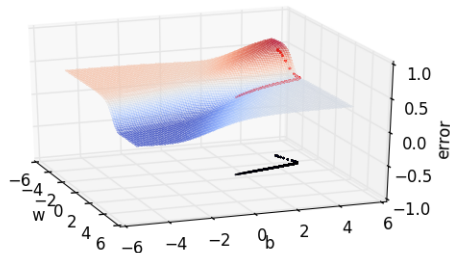
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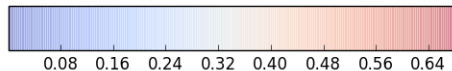
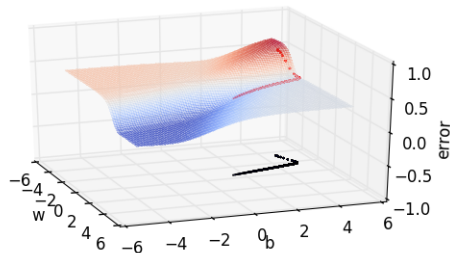
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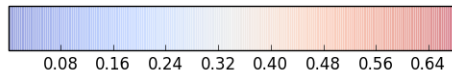
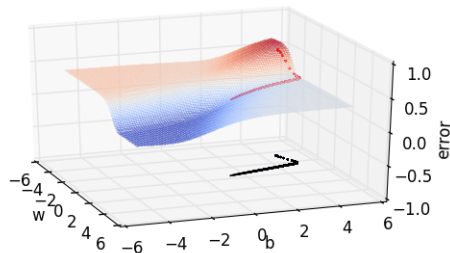
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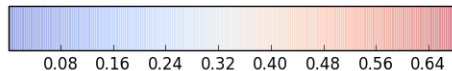
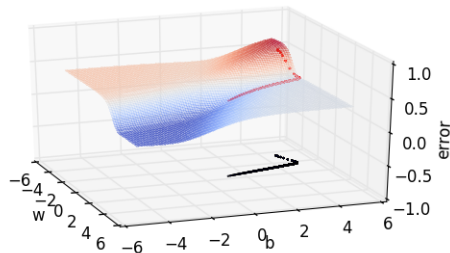
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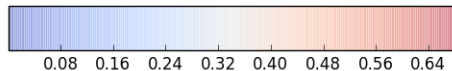
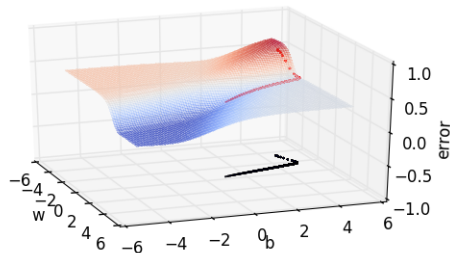
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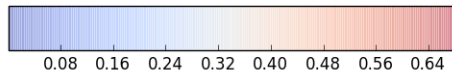
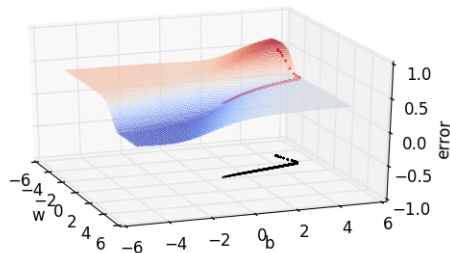
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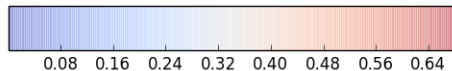
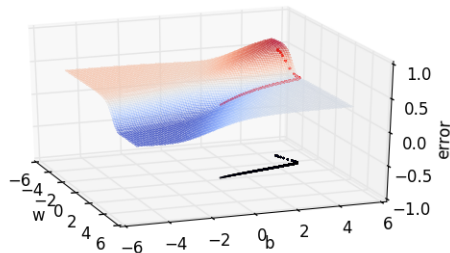
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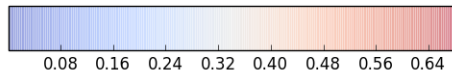
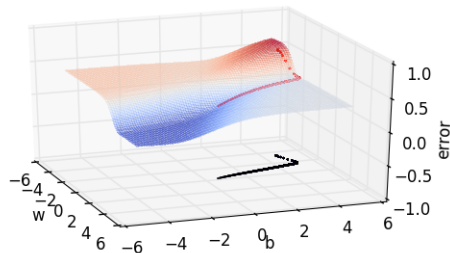
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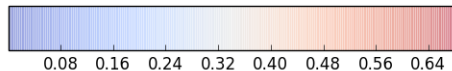
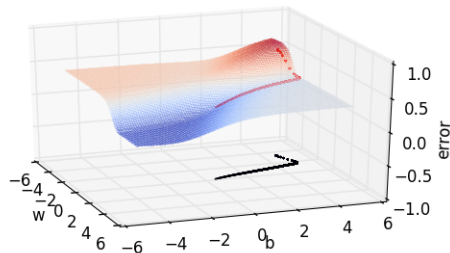
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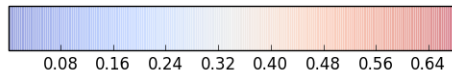
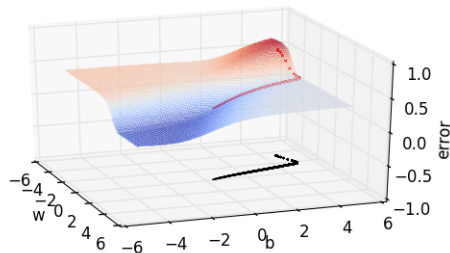
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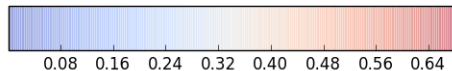
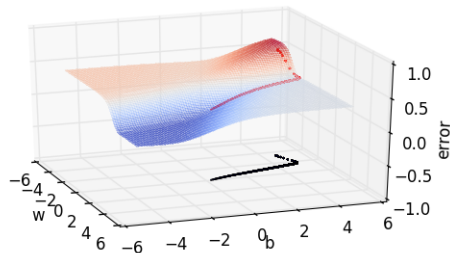
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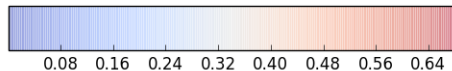
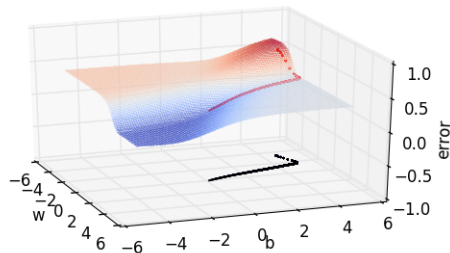
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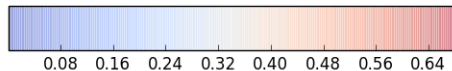
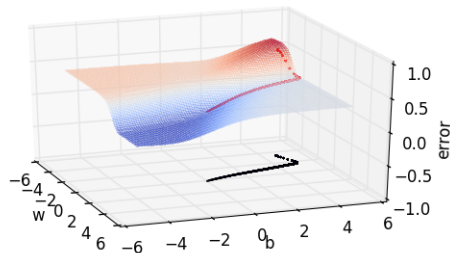
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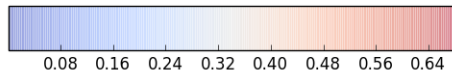
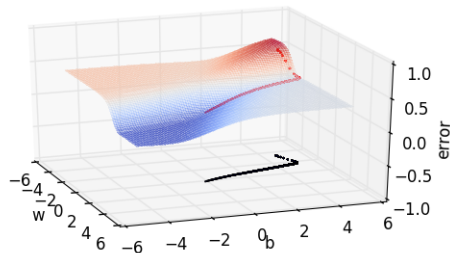
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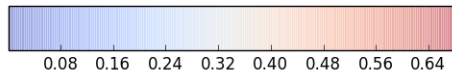
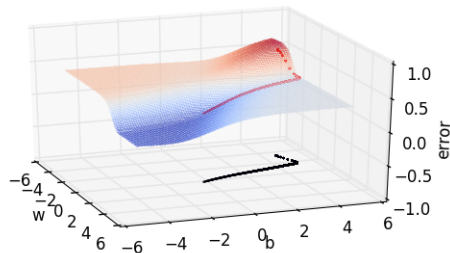
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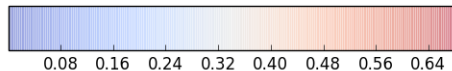
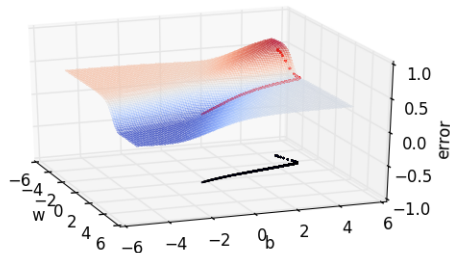
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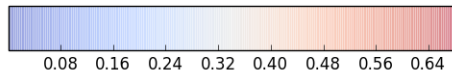
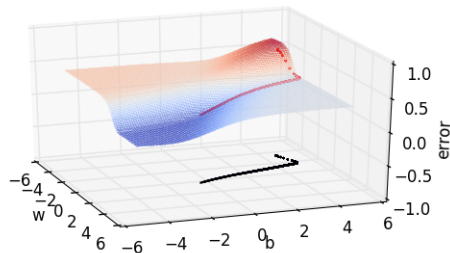
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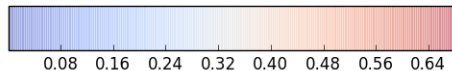
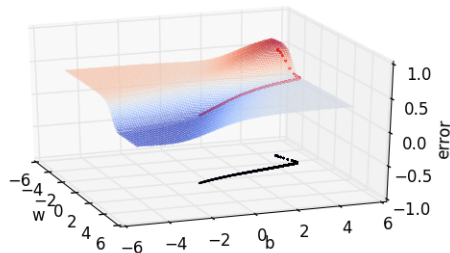
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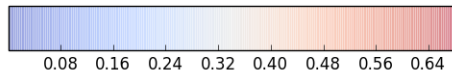
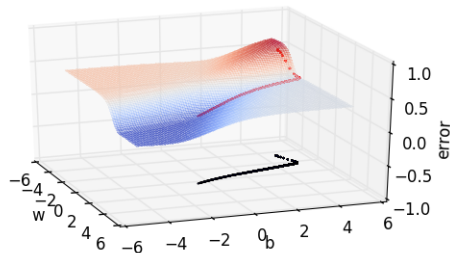
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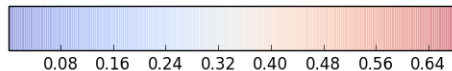
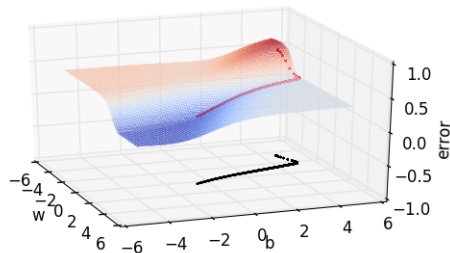
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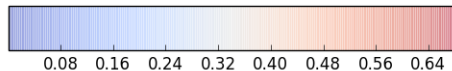
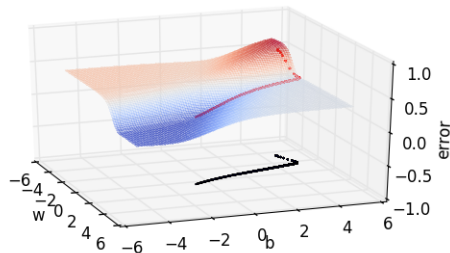
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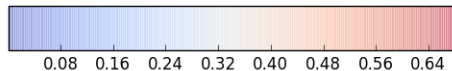
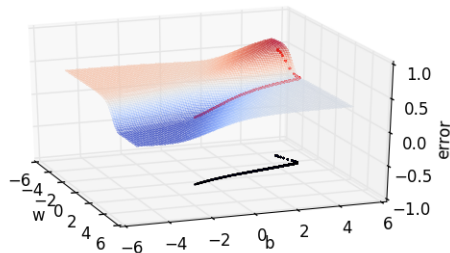
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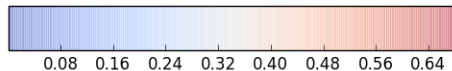
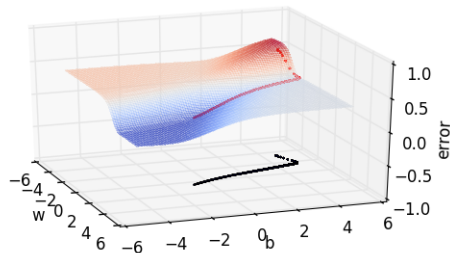
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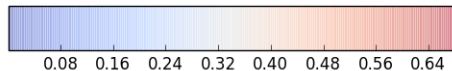
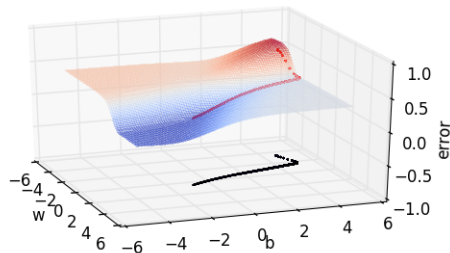
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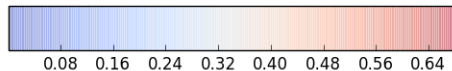
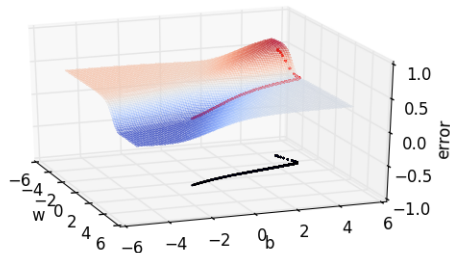
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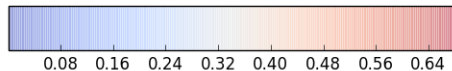
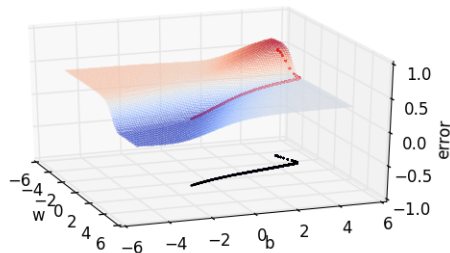
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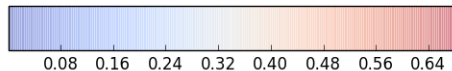
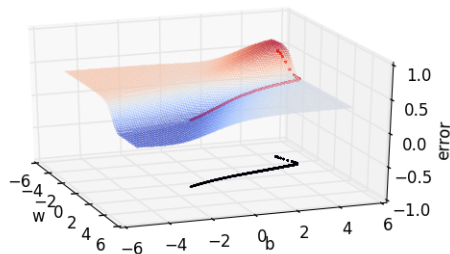
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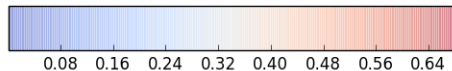
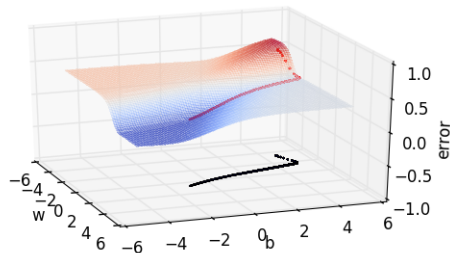
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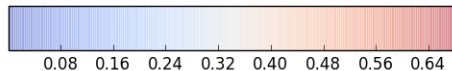
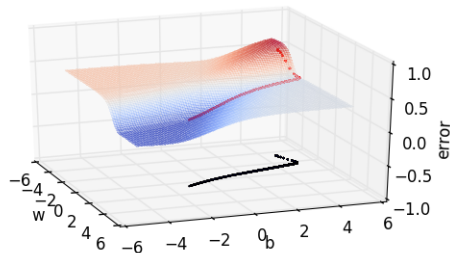
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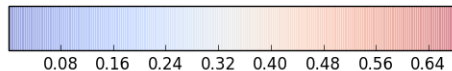
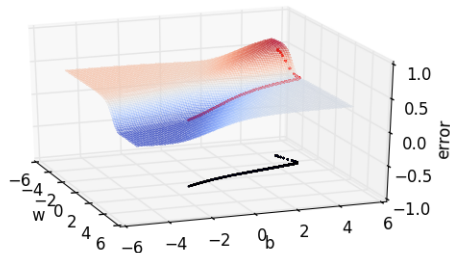
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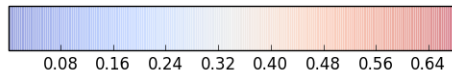
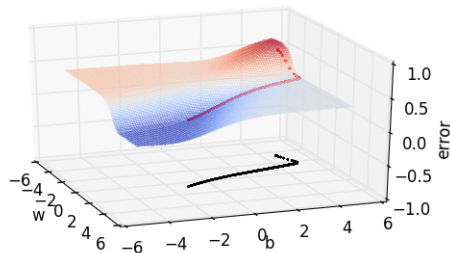
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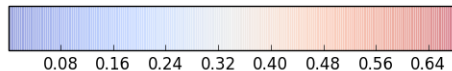
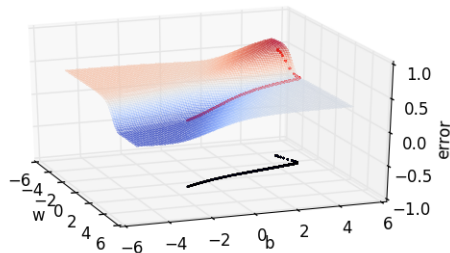
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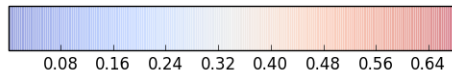
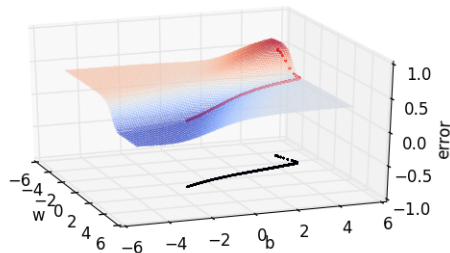
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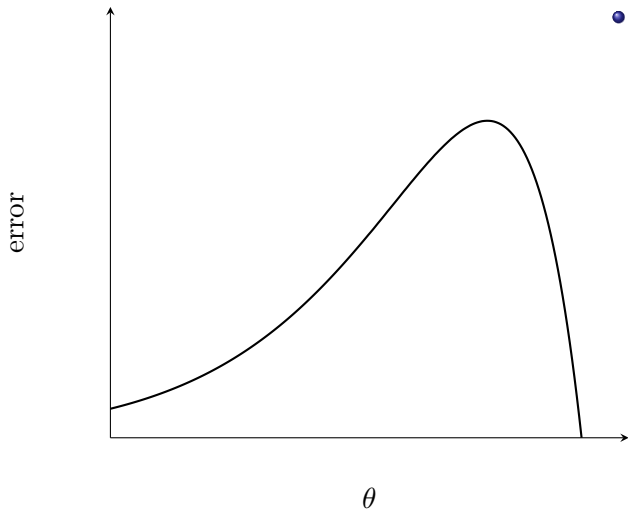


Module 5.3 : Contours

- *Visualizing things in 3d can sometimes become a bit cumbersome*

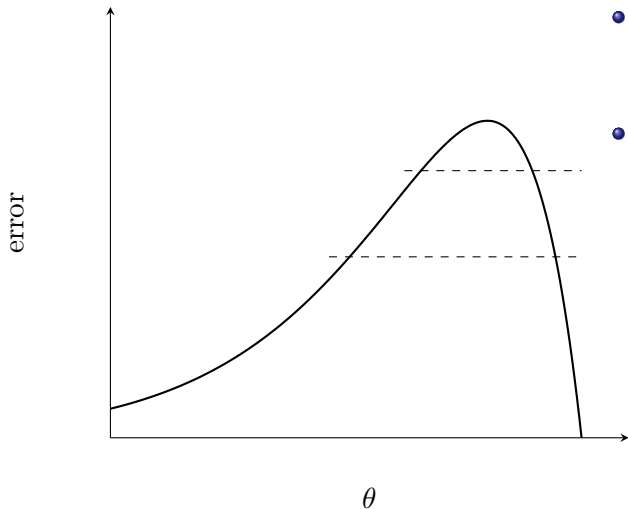
- *Visualizing things in 3d can sometimes become a bit cumbersome*
- *Can we do a 2d visualization of this traversal along the error surface*

- *Visualizing things in 3d can sometimes become a bit cumbersome*
- *Can we do a 2d visualization of this traversal along the error surface*
- *Yes, let's take a look at something known as contours*



- Suppose I take horizontal slices of this error surface at regular intervals along the vertical axis

Figure: Front view of a 3d error surface



- Suppose I take horizontal slices of this error surface at regular intervals along the vertical axis
- How would this look from the top-view ?

Figure: Front view of a 3d error surface

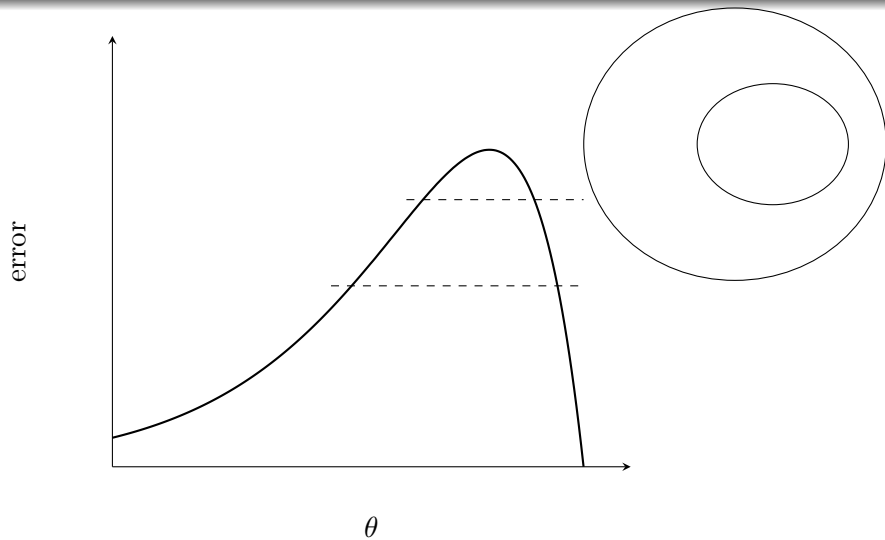


Figure: Front view of a 3d error surface

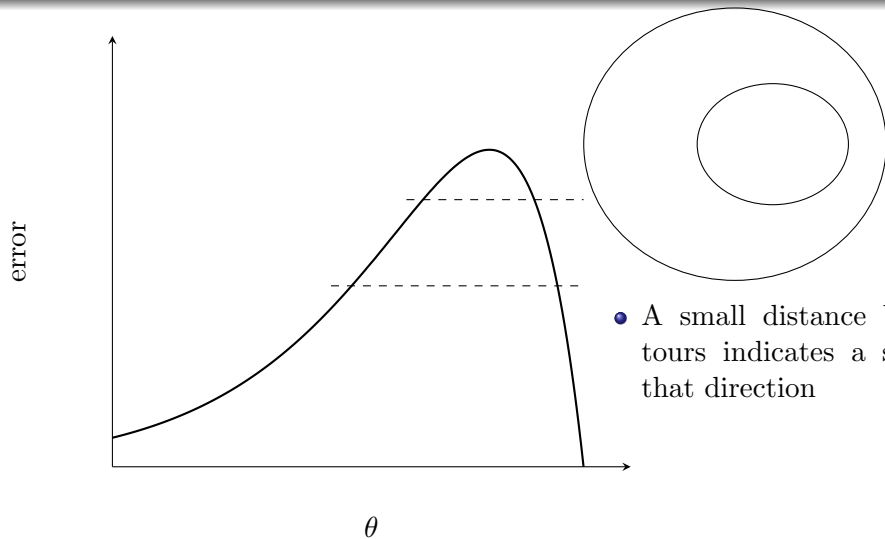


Figure: Front view of a 3d error surface

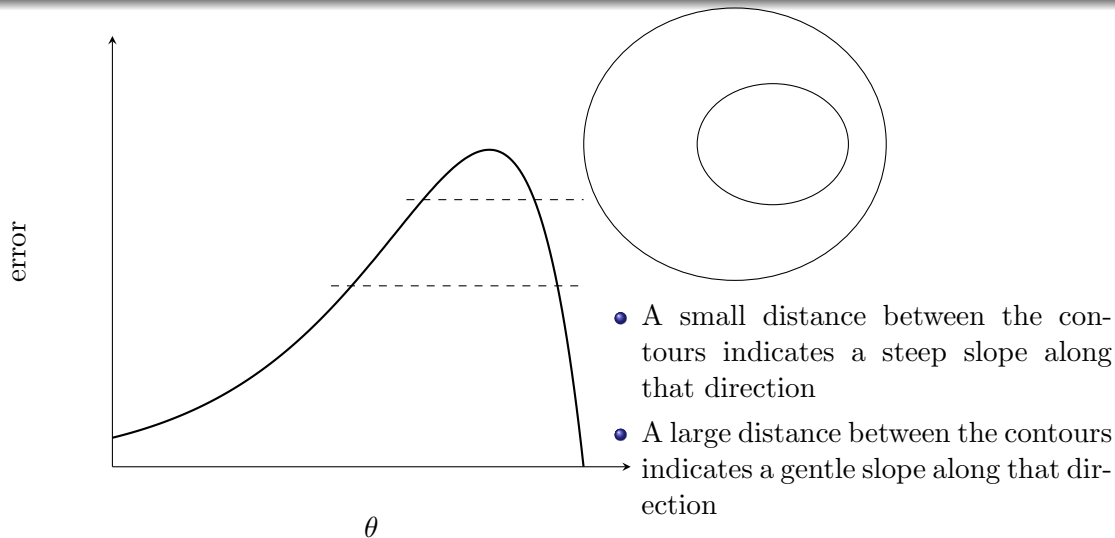
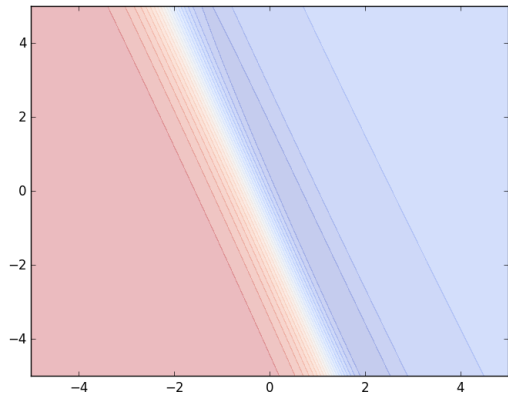
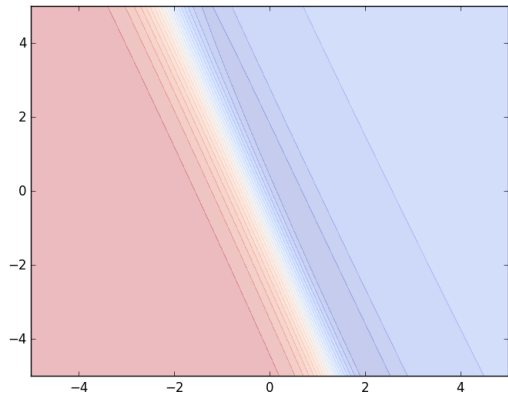


Figure: Front view of a 3d error surface

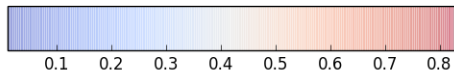
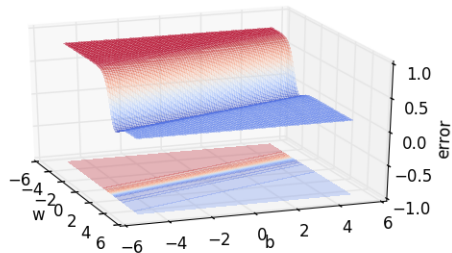
- *Just to ensure that we understand this properly let us do a few exercises ...*

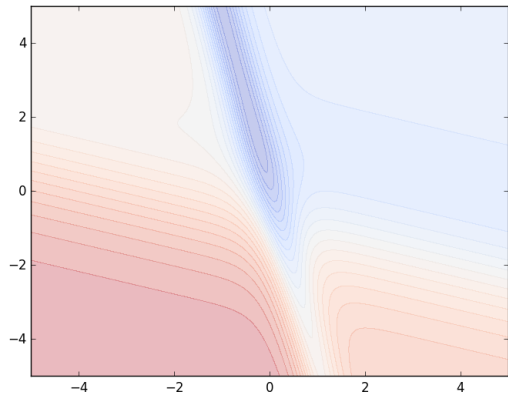


Guess the 3d surface

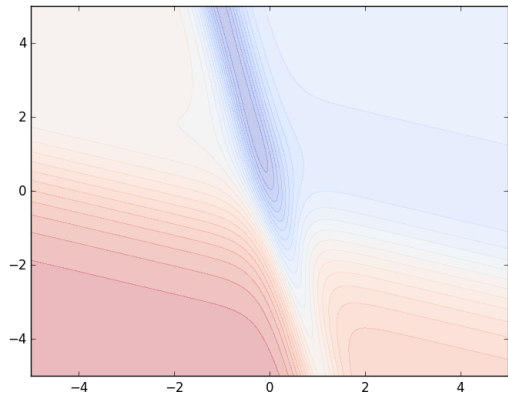


Guess the 3d surface

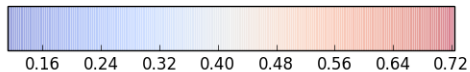
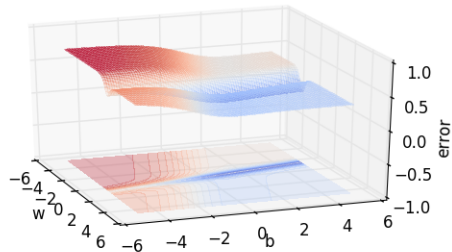


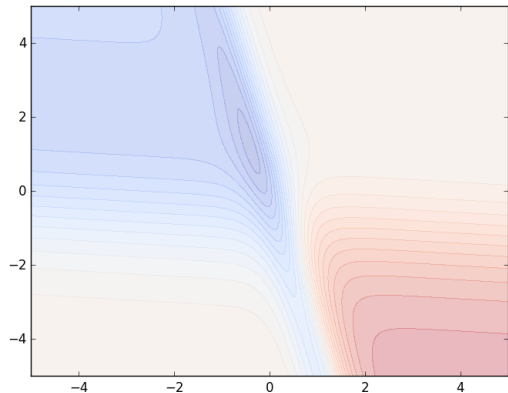


Guess the 3d surface

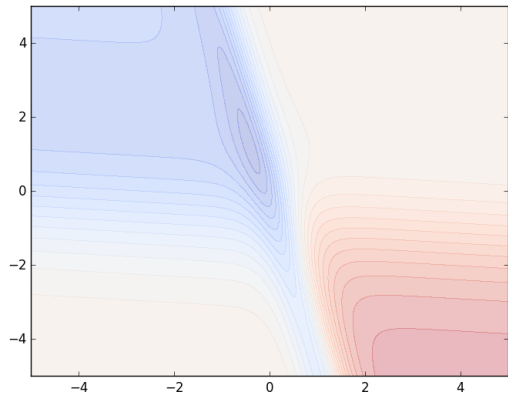


Guess the 3d surface

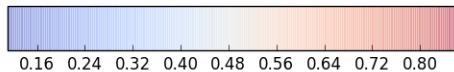
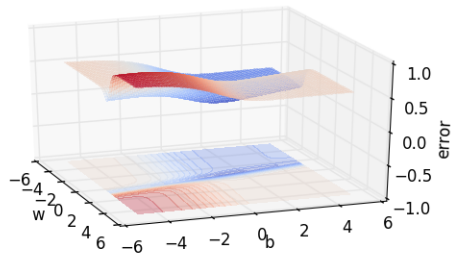




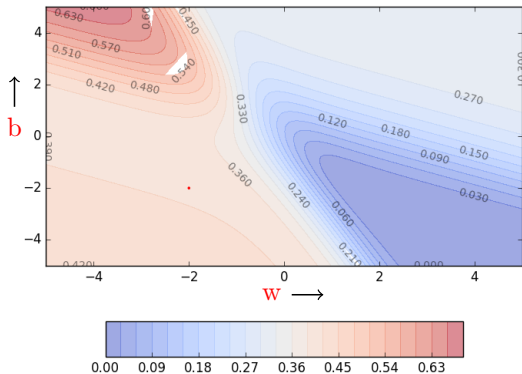
Guess the 3d surface



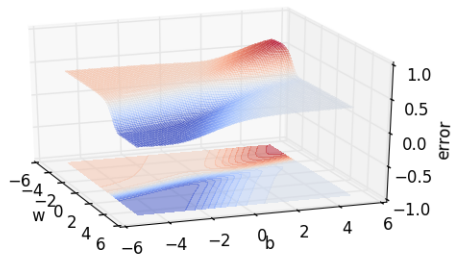
Guess the 3d surface

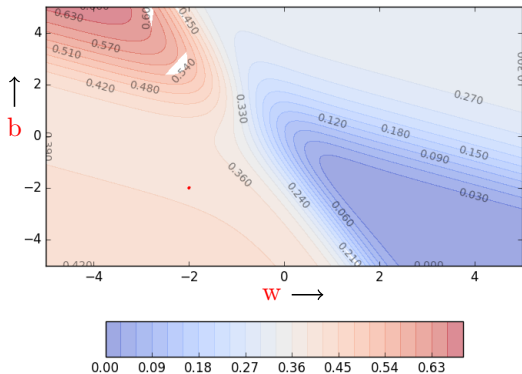


- *Now that we know what are contour maps and how to read them let us go back to our toy example and visualize gradient descent from the point of view of contours...*

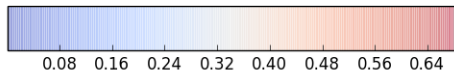
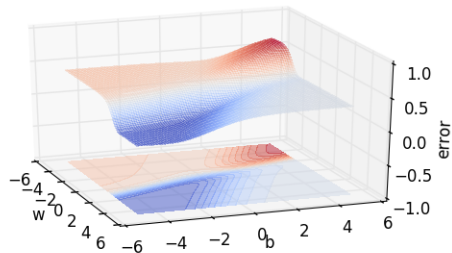


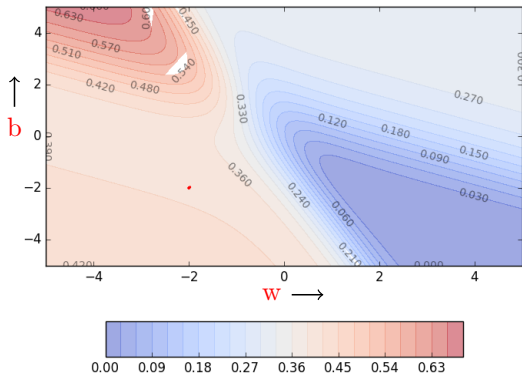
Gradient descent on the error surface



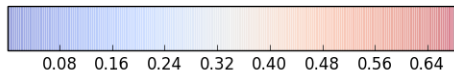
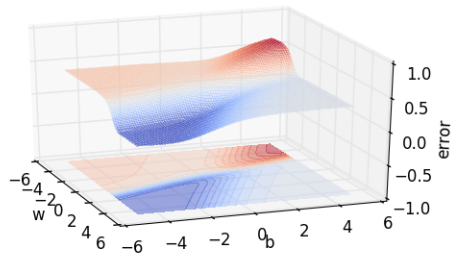


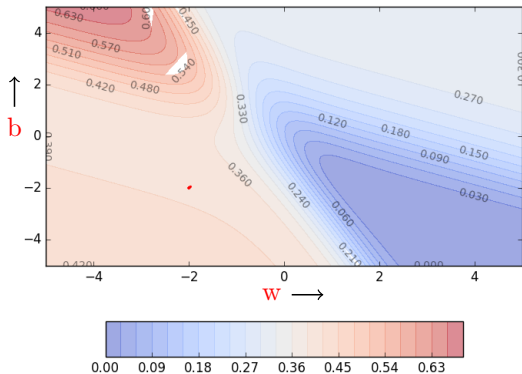
Gradient descent on the error surface



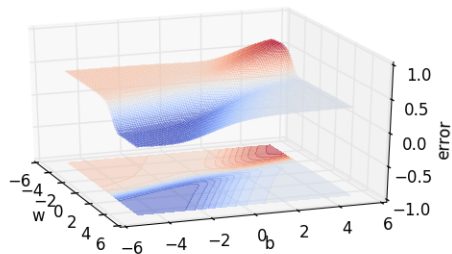


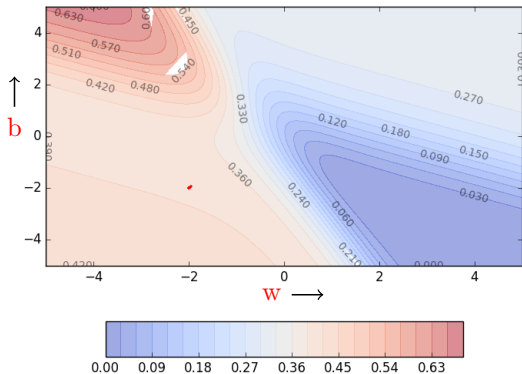
Gradient descent on the error surface



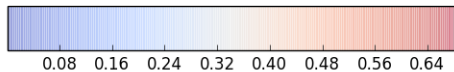
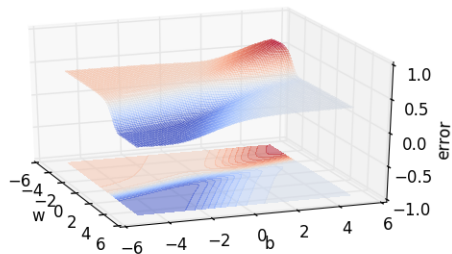


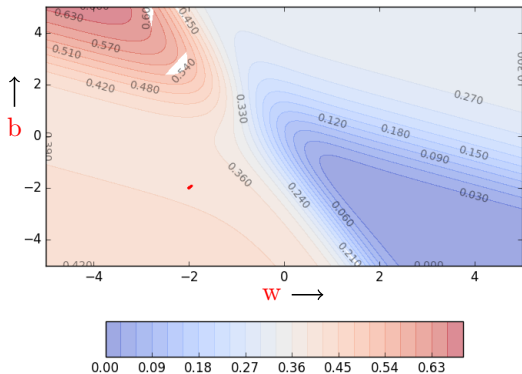
Gradient descent on the error surface



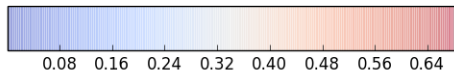
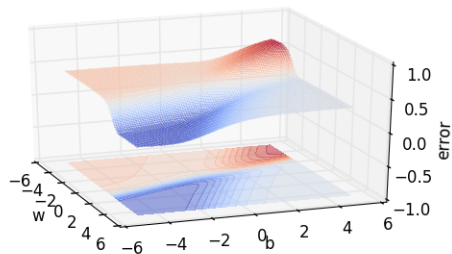


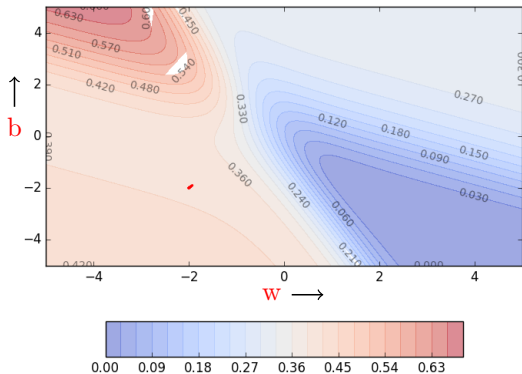
Gradient descent on the error surface



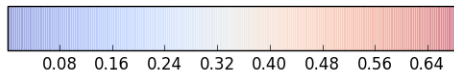
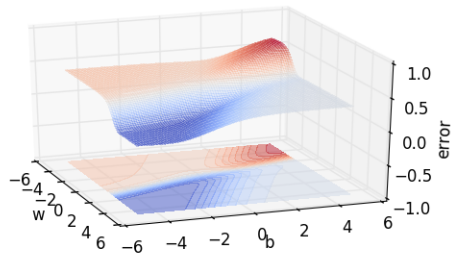


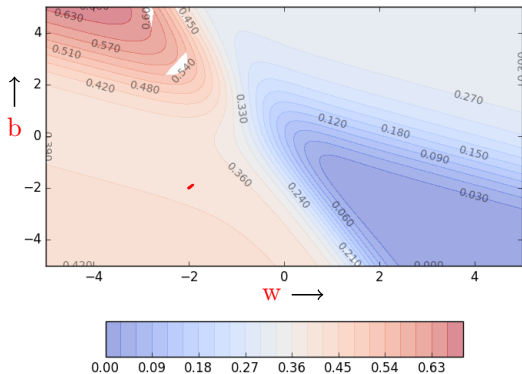
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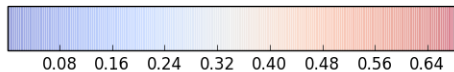
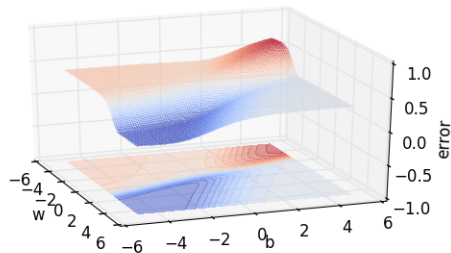


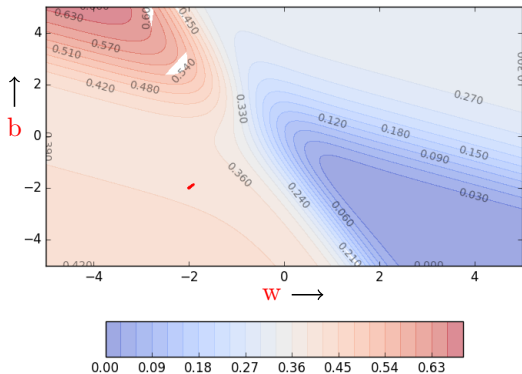
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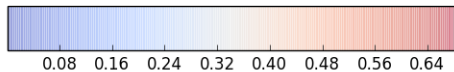
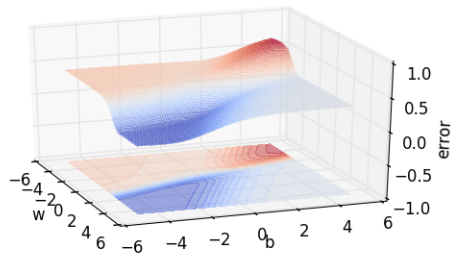


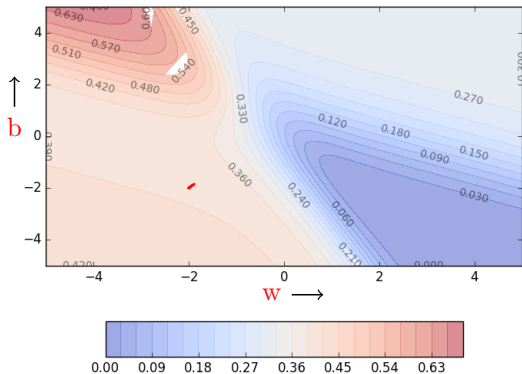
Gradient descent on the error surface



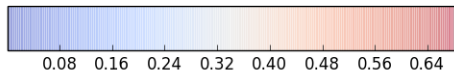
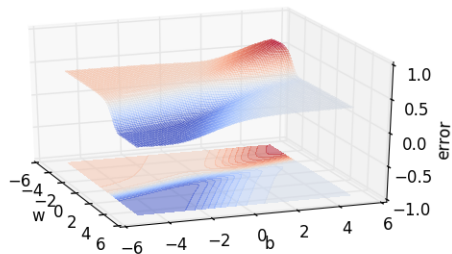


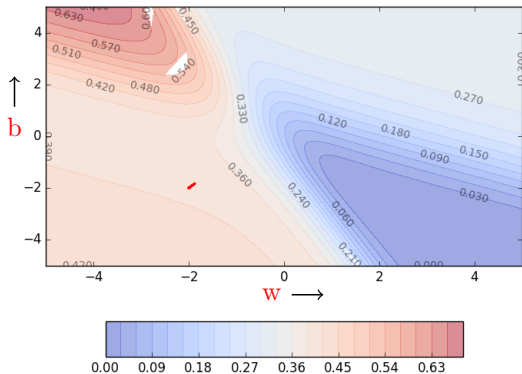
Gradient descent on the error surface



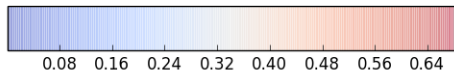
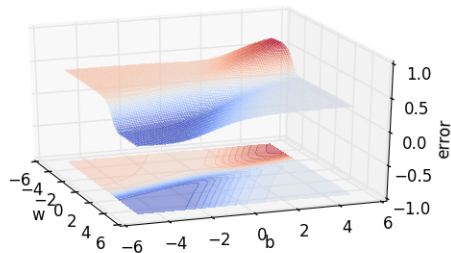


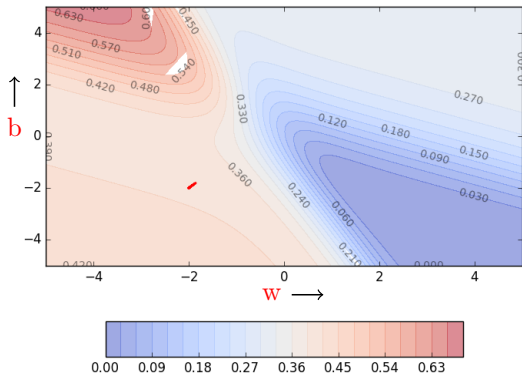
Gradient descent on the error surface



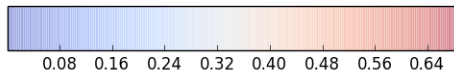
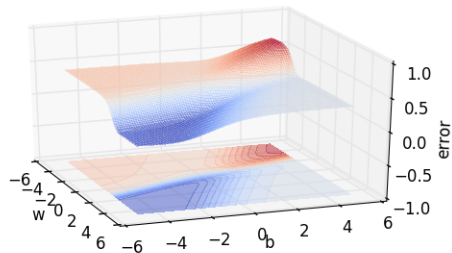


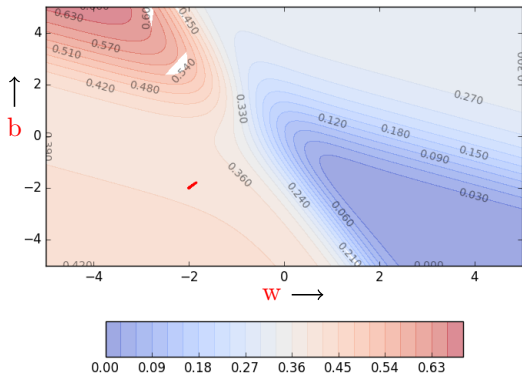
Gradient descent on the error surface



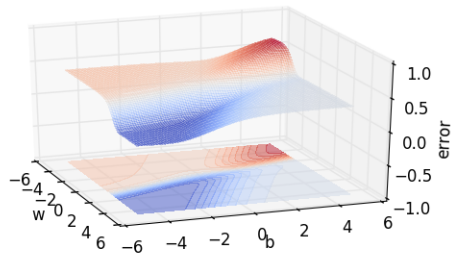


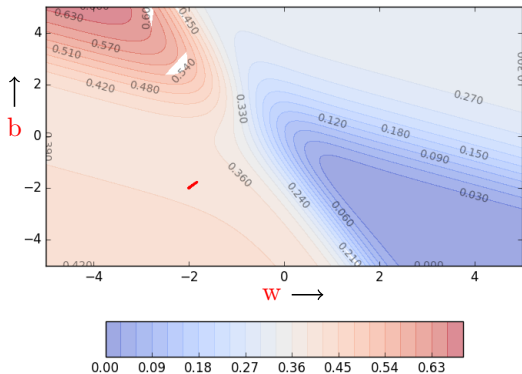
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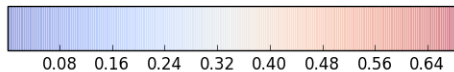
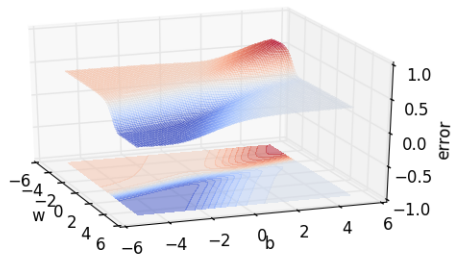


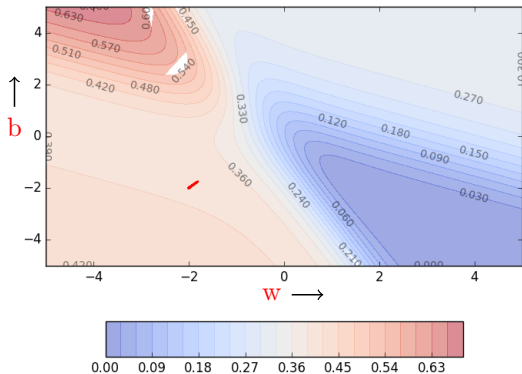
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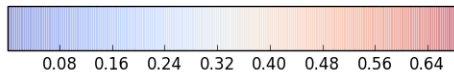
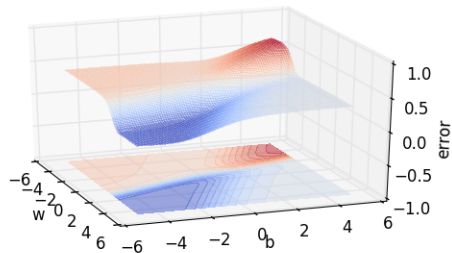


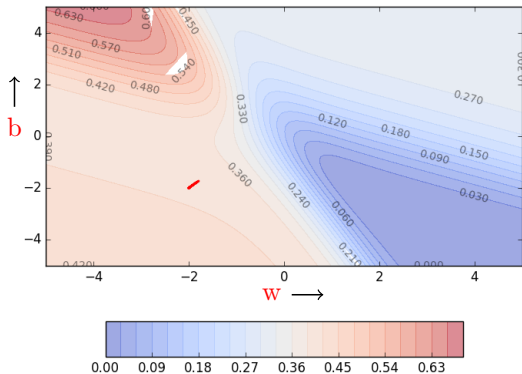
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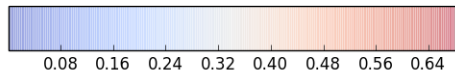
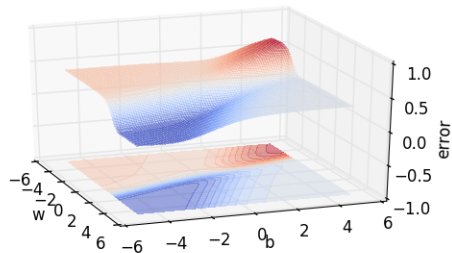


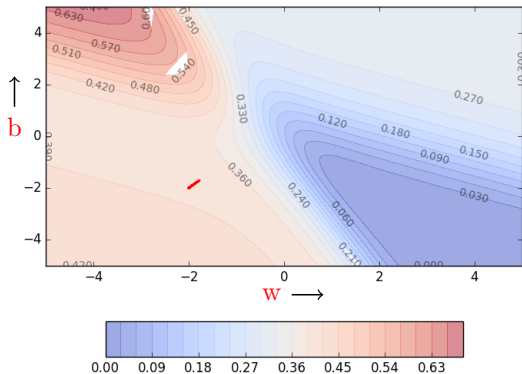
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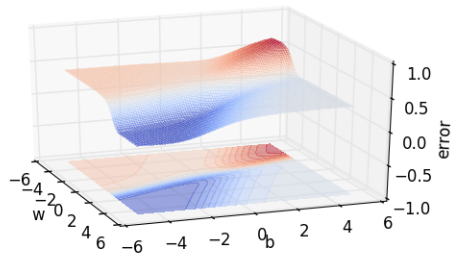


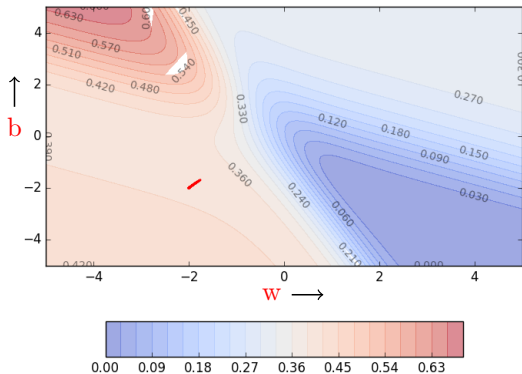
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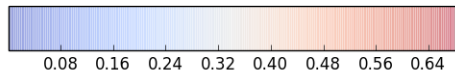
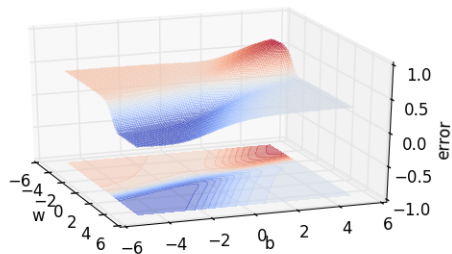


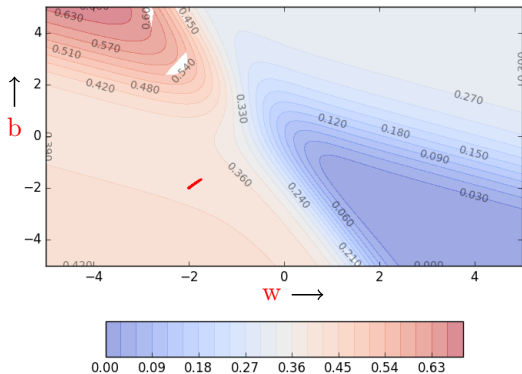
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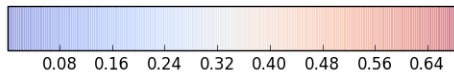
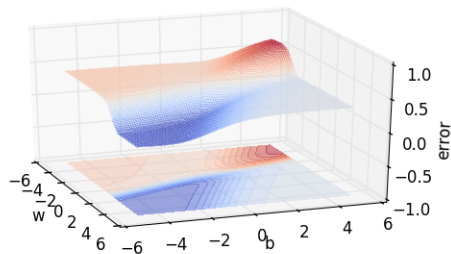


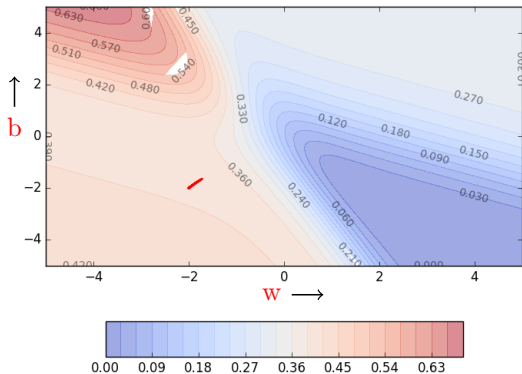
Gradient descent on the error surface



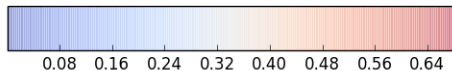
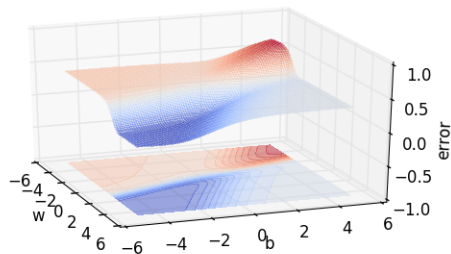


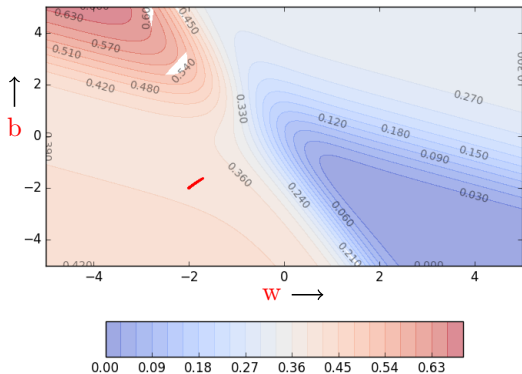
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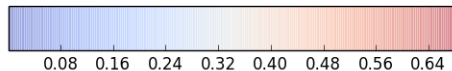
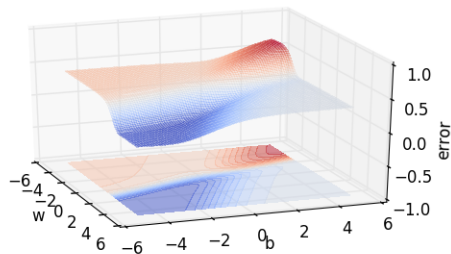


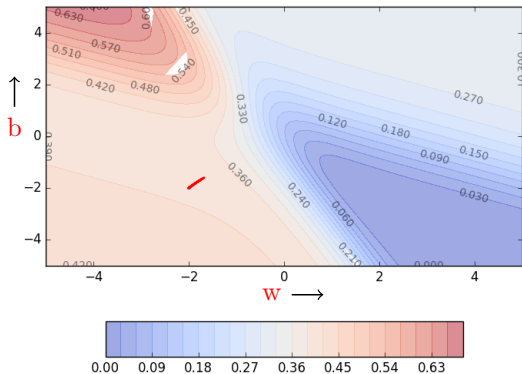
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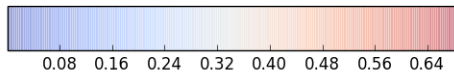
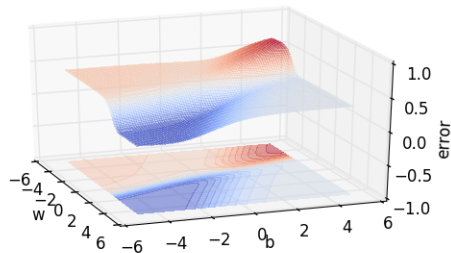


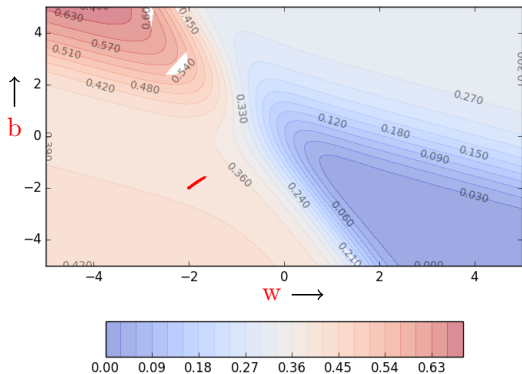
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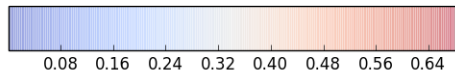
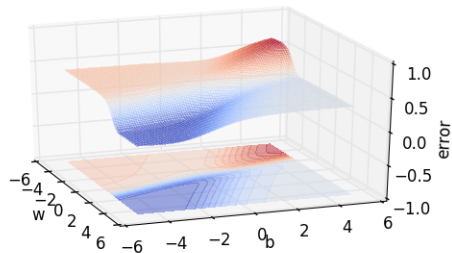


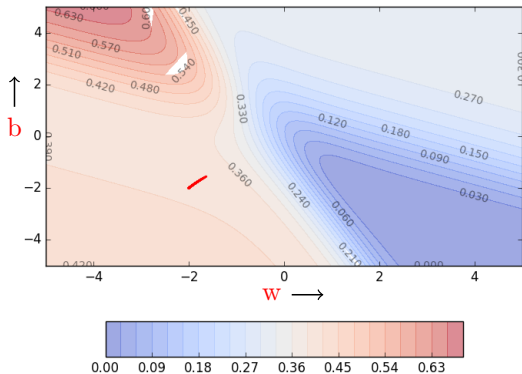
Gradient descent on the error surface



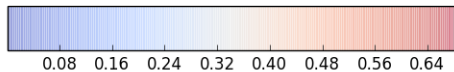
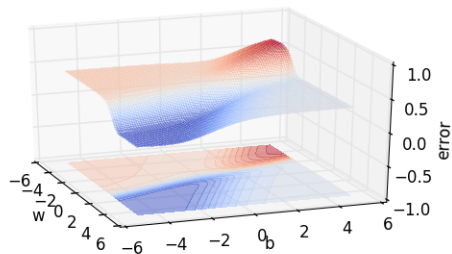


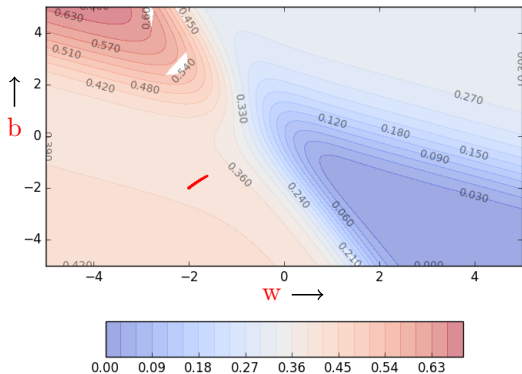
Gradient descent on the error surface



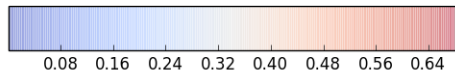
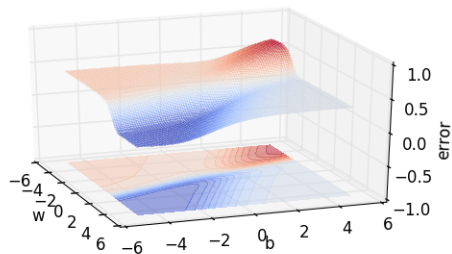


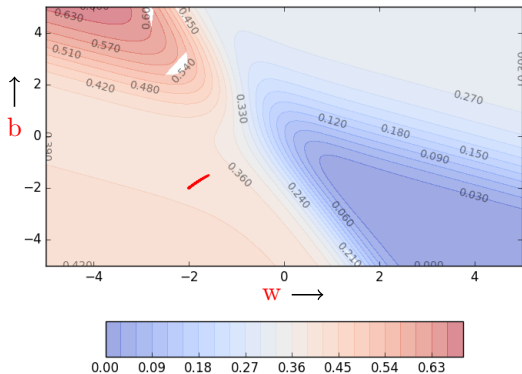
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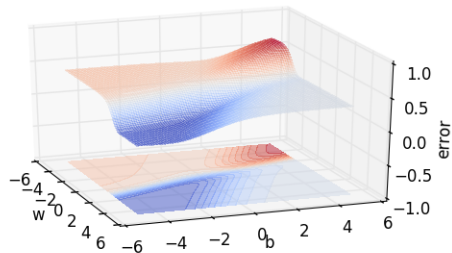


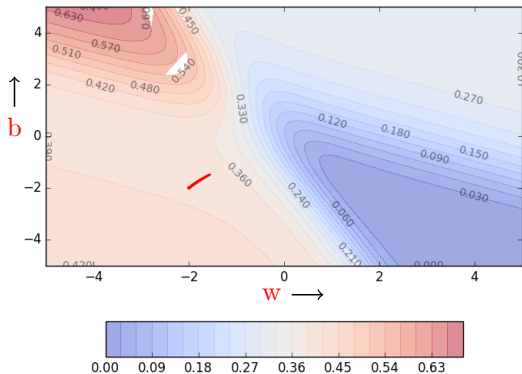
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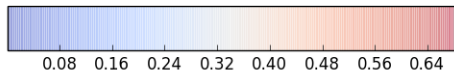
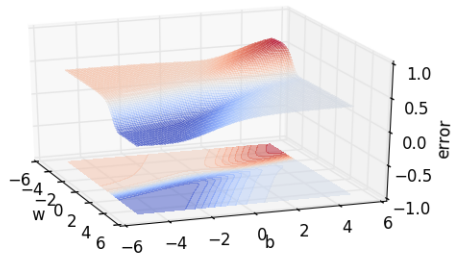


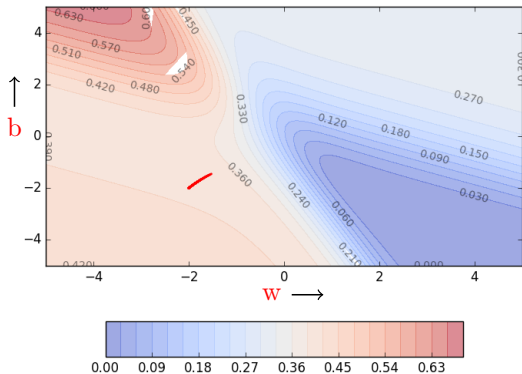
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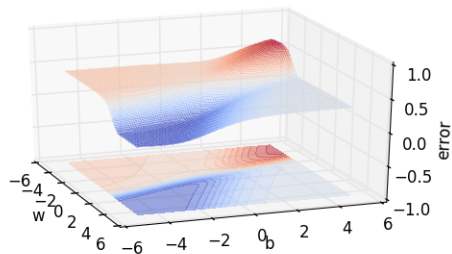


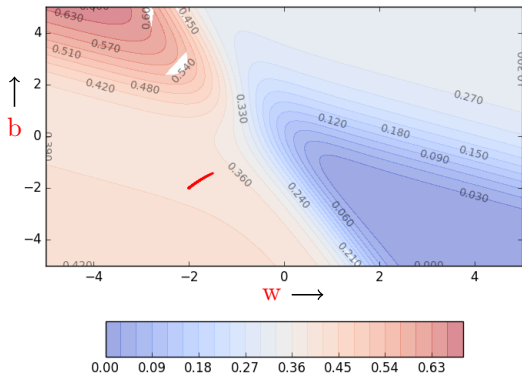
Gradient descent on the error surface



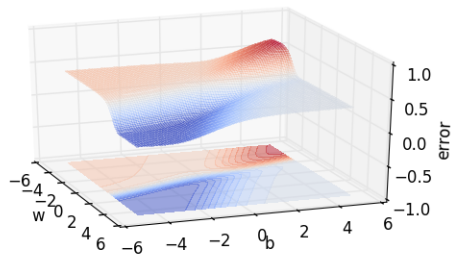


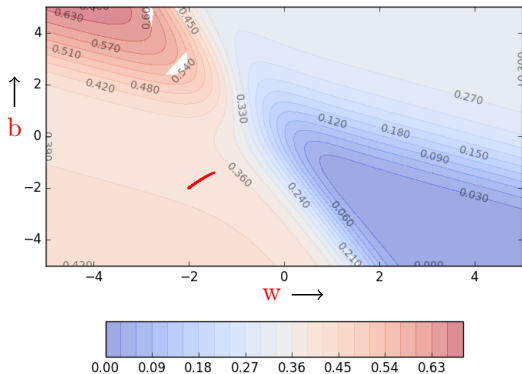
Gradient descent on the error surface



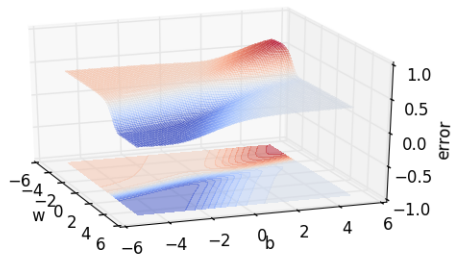


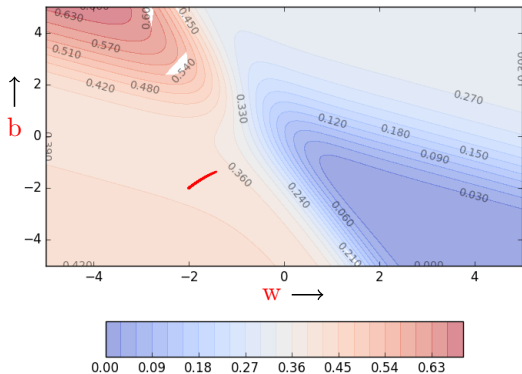
Gradient descent on the error surface



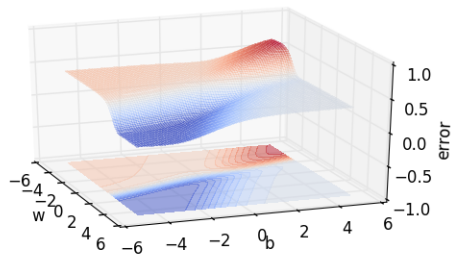


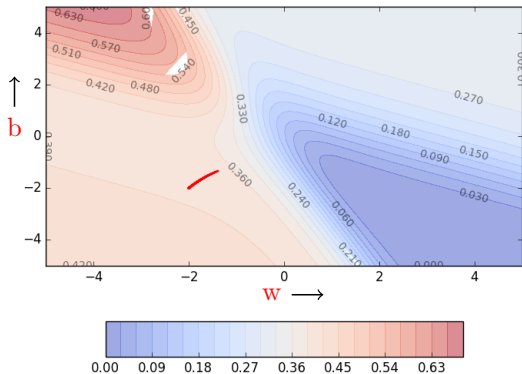
Gradient descent on the error surface



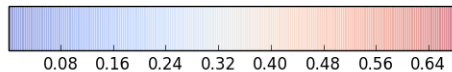
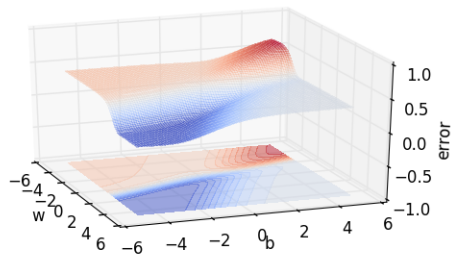


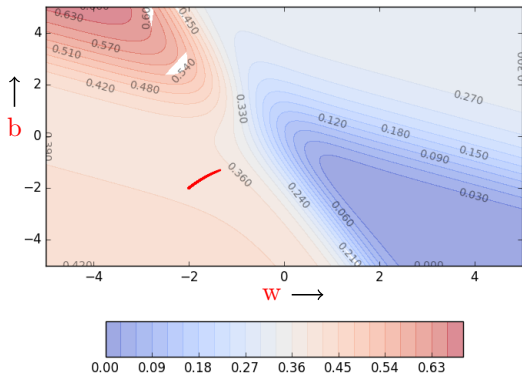
Gradient descent on the error surface



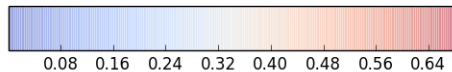
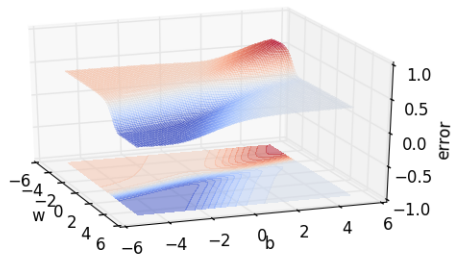


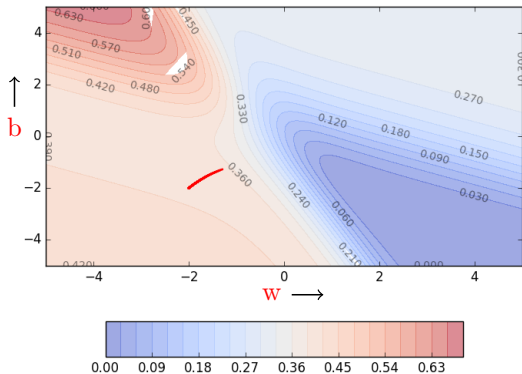
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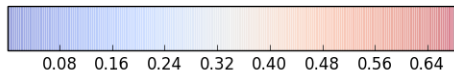
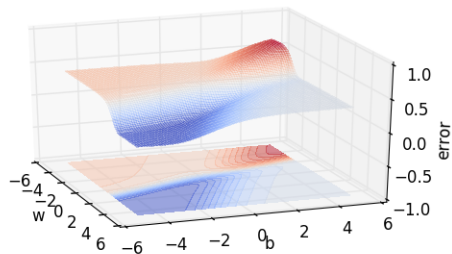


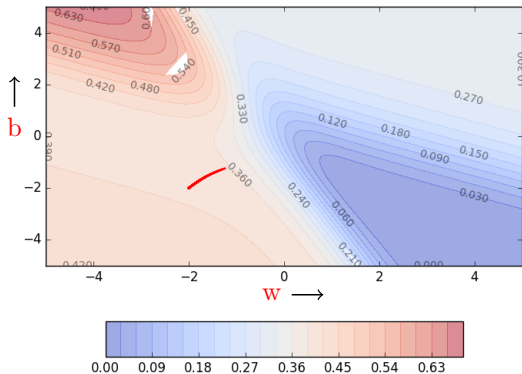
Gradient descent on the error surface



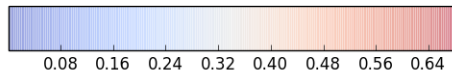
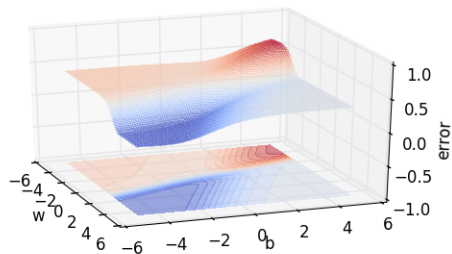


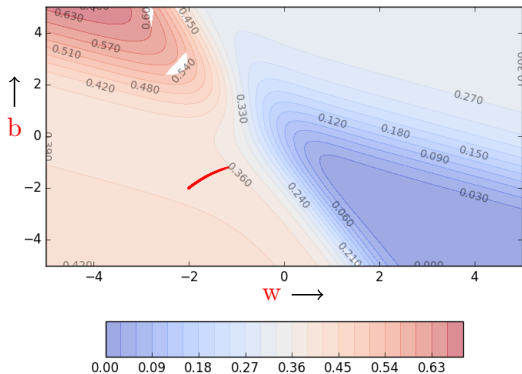
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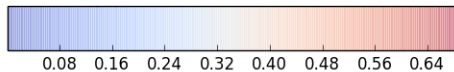
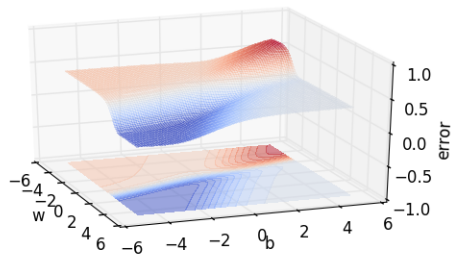


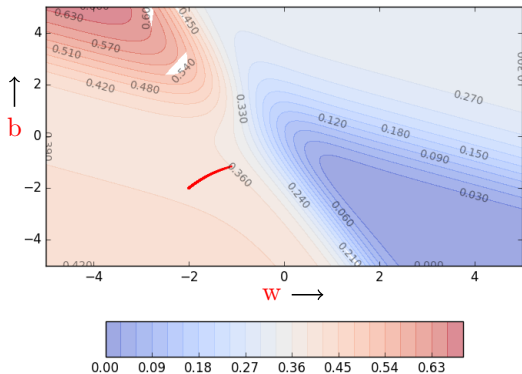
Gradient descent on the error surface



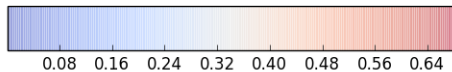
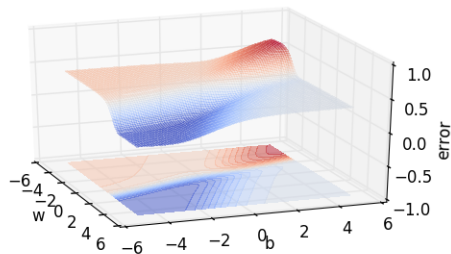


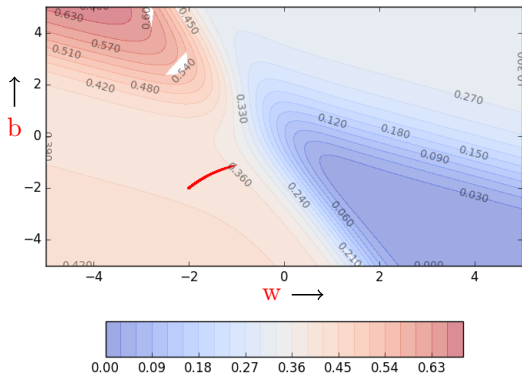
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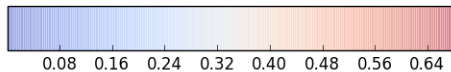
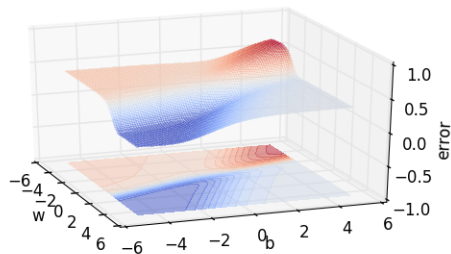


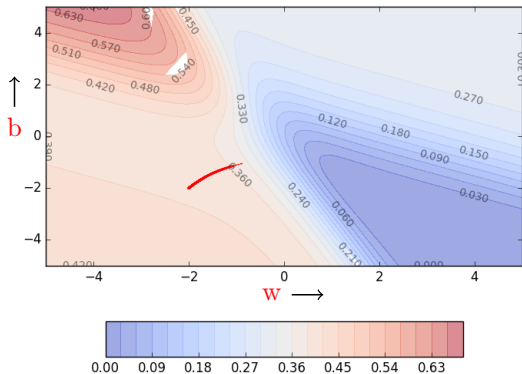
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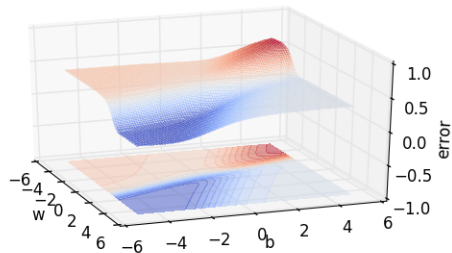


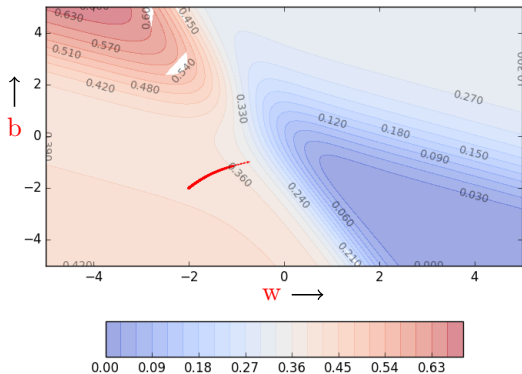
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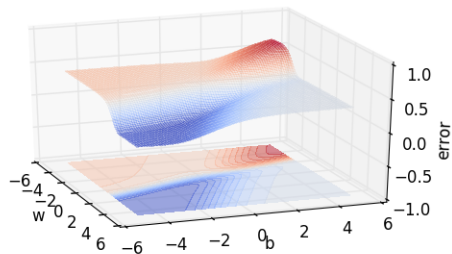


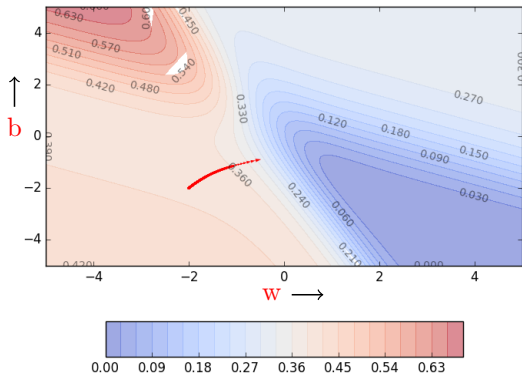
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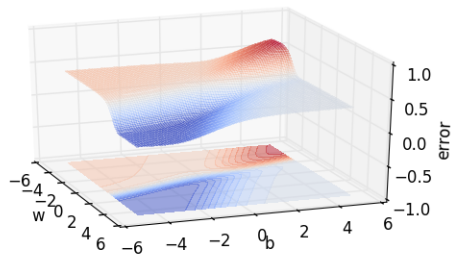


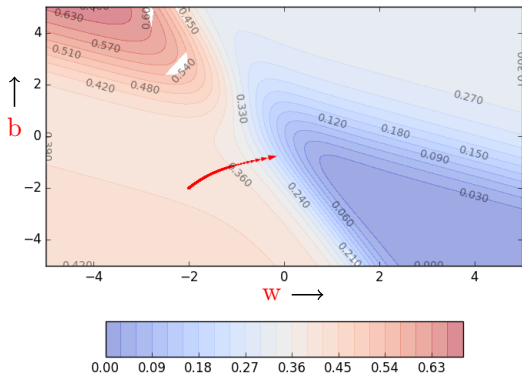
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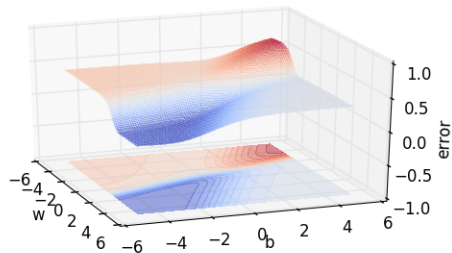


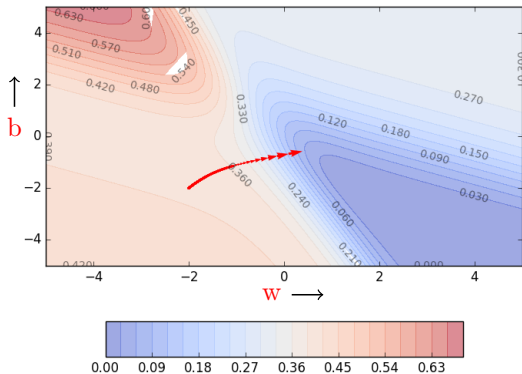
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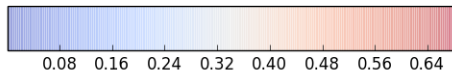
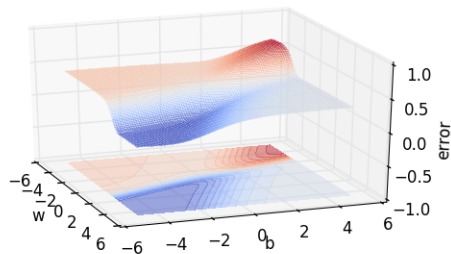


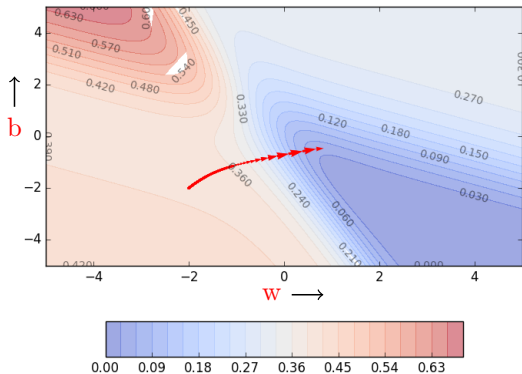
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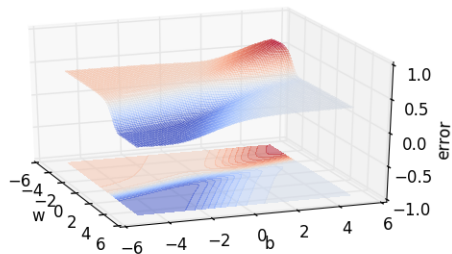


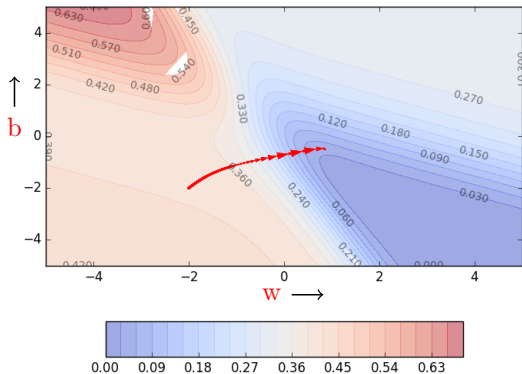
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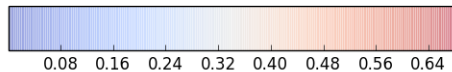
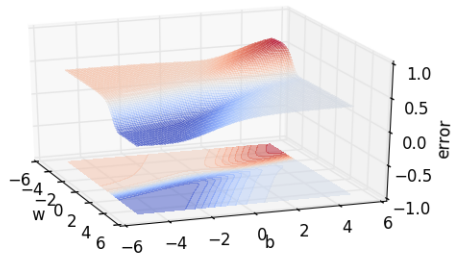


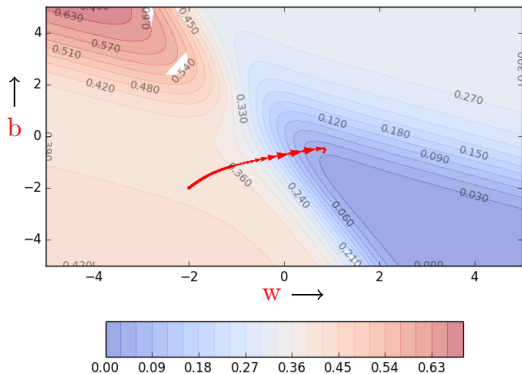
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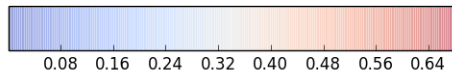
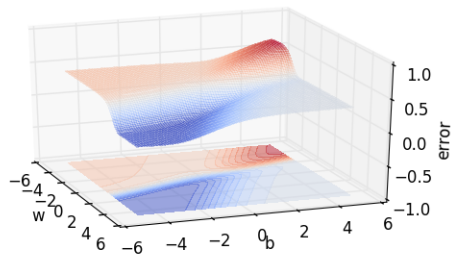


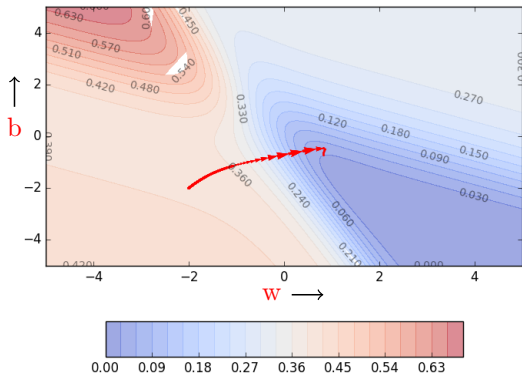
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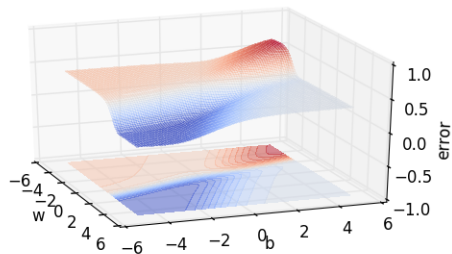


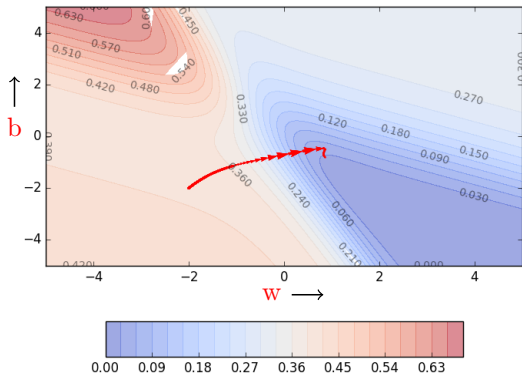
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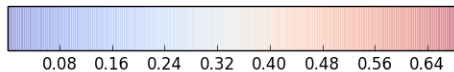
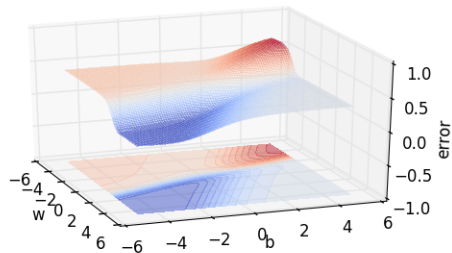


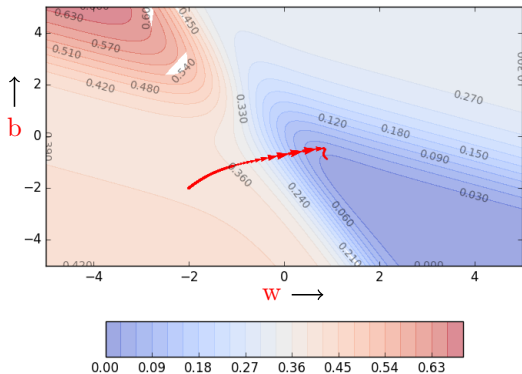
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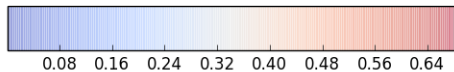
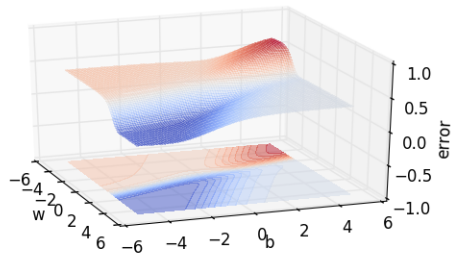


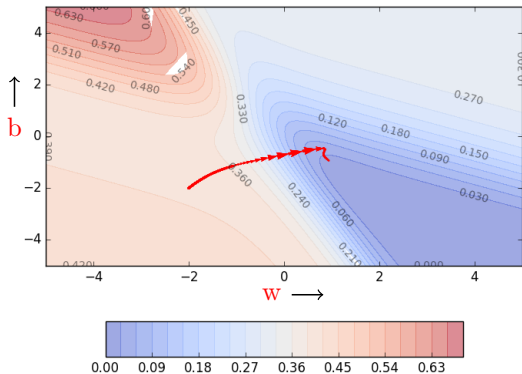
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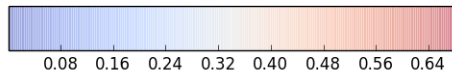
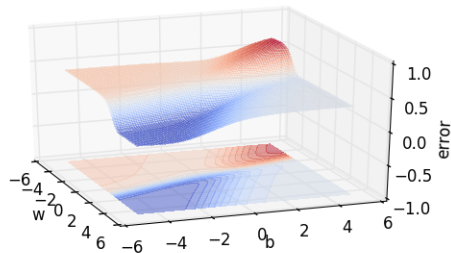


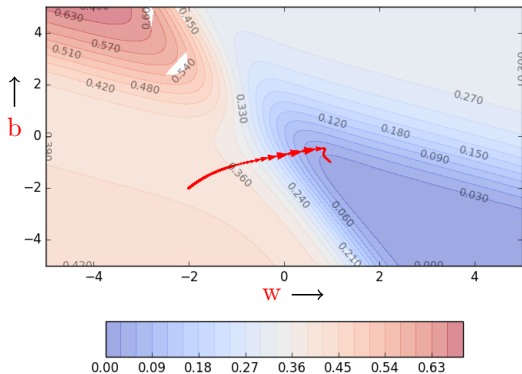
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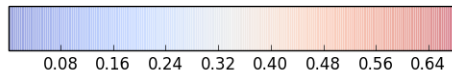
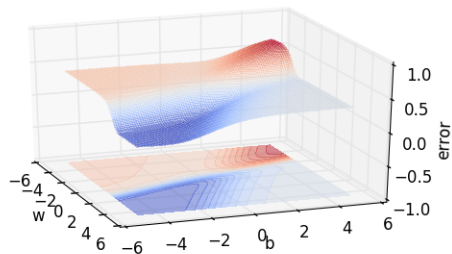


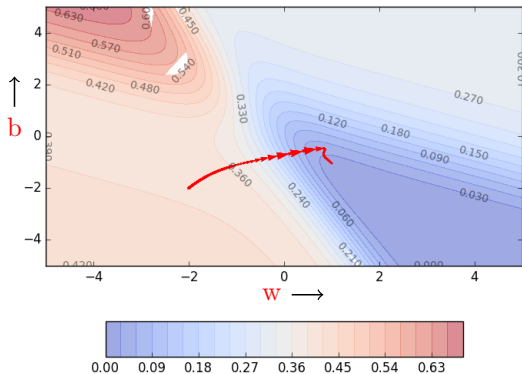
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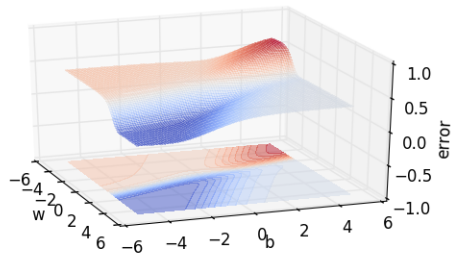


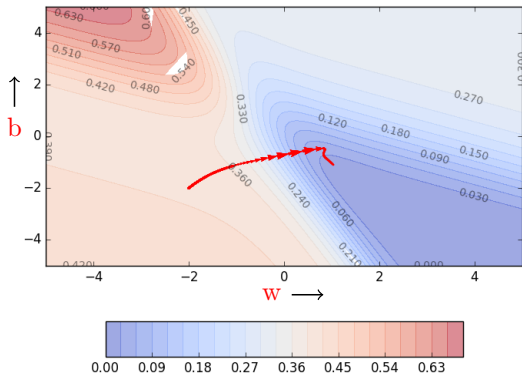
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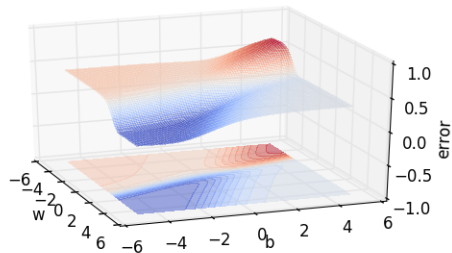


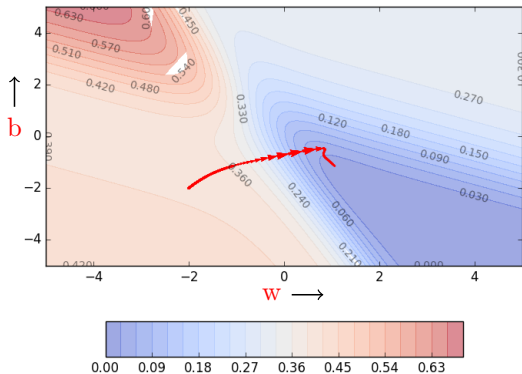
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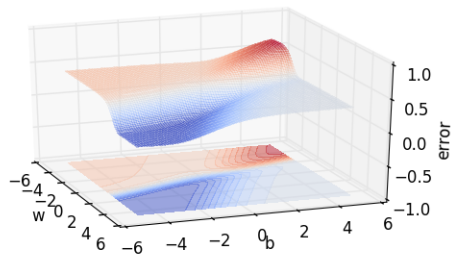


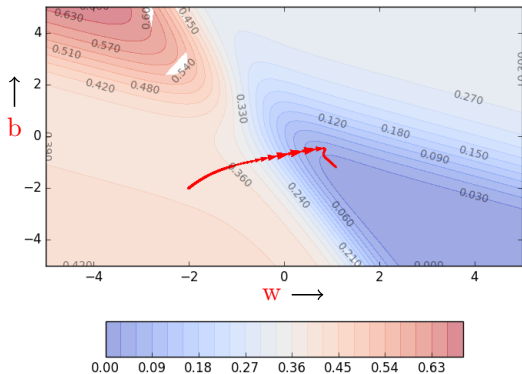
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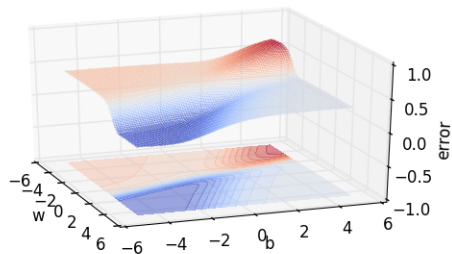


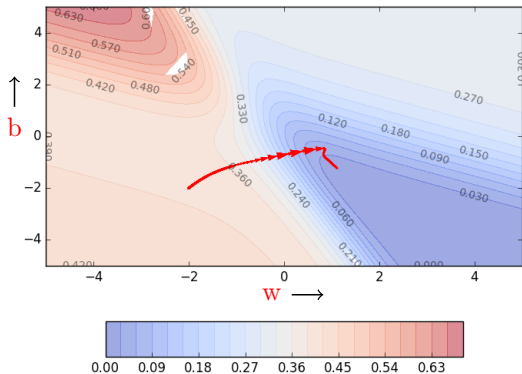
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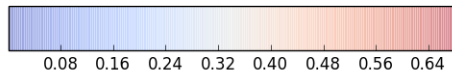
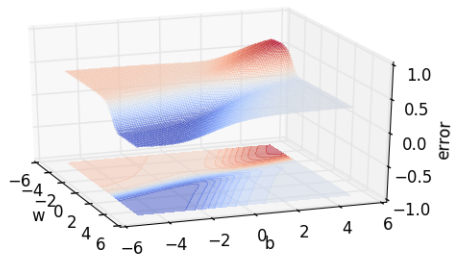


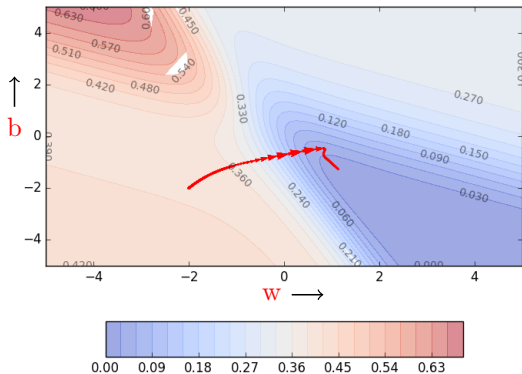
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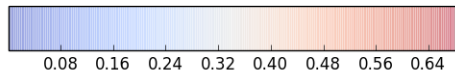
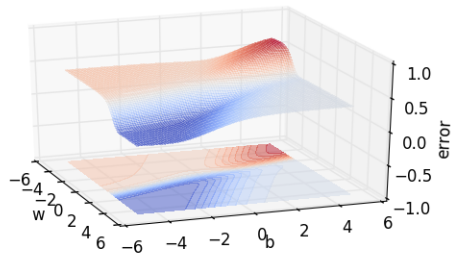


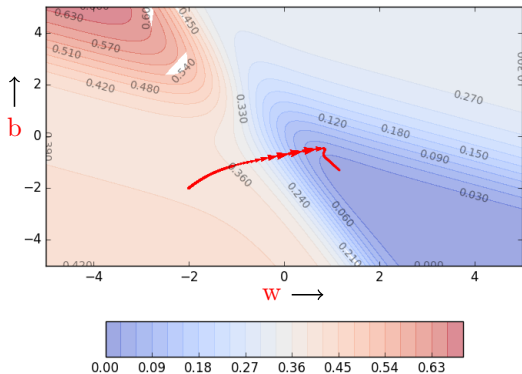
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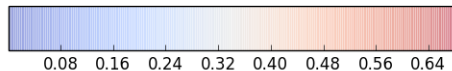
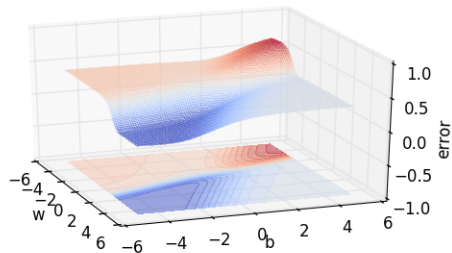


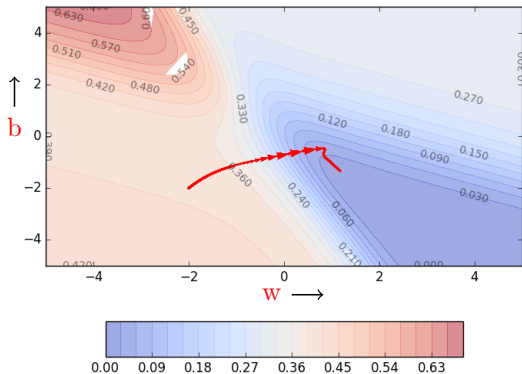
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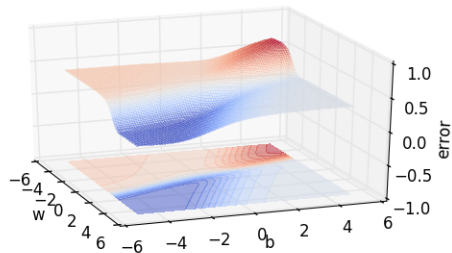


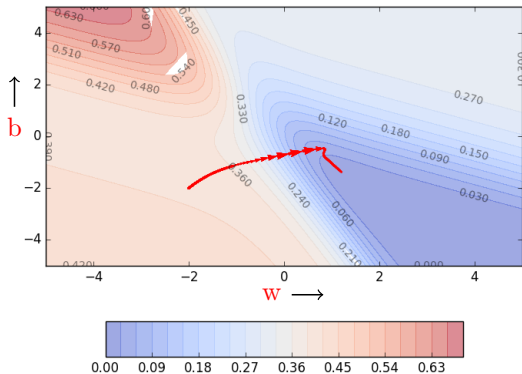
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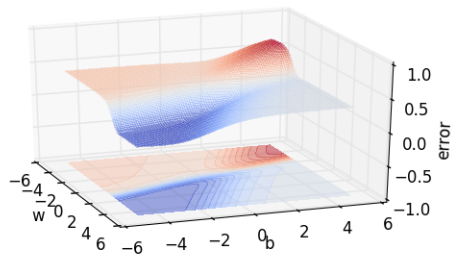


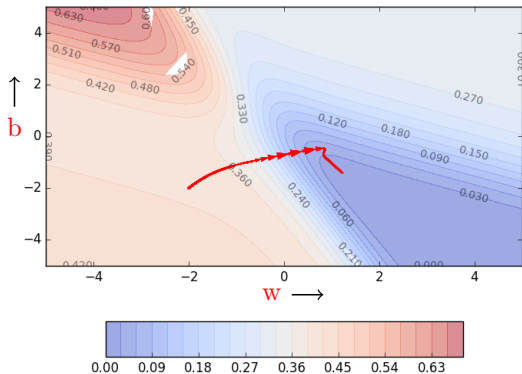
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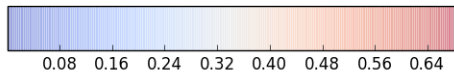
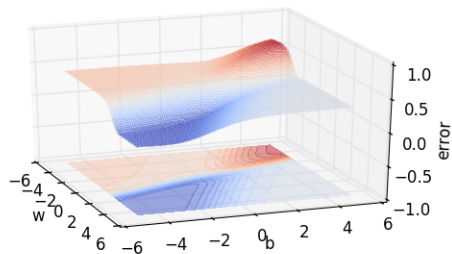


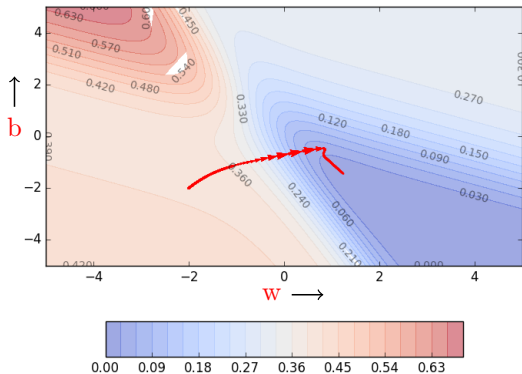
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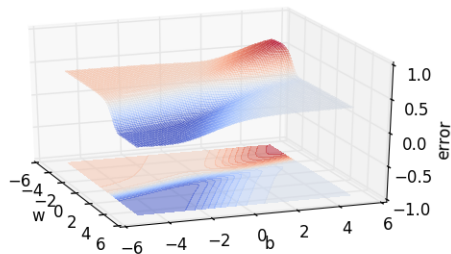


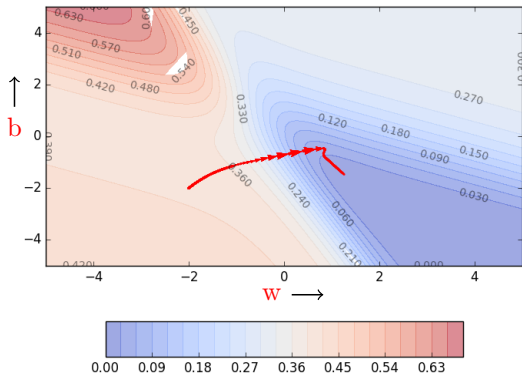
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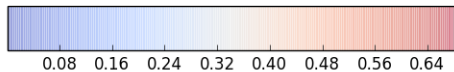
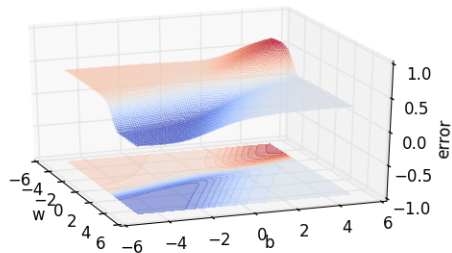


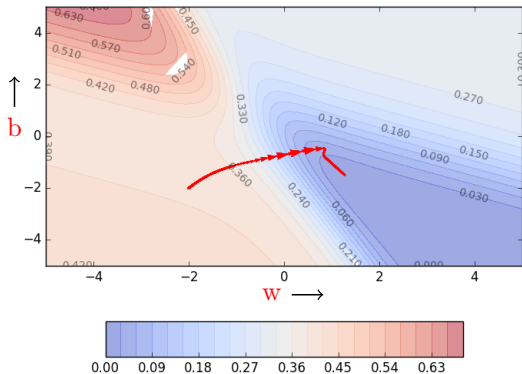
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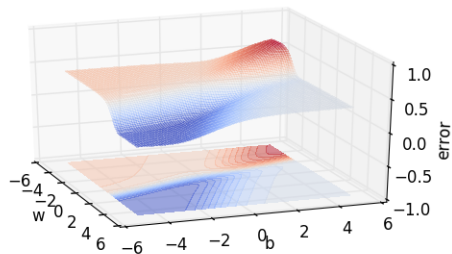


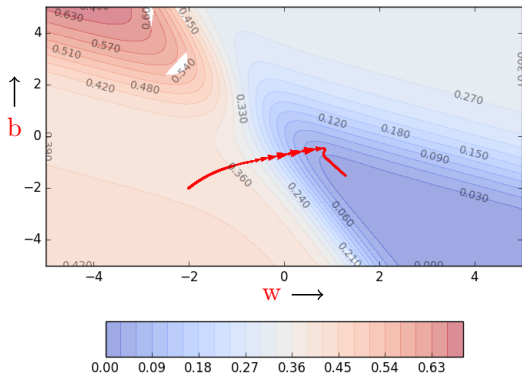
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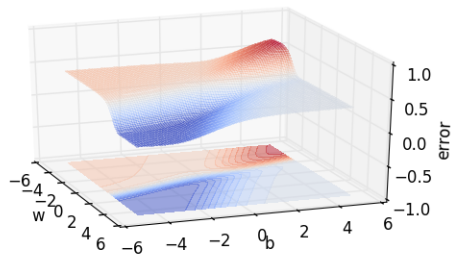


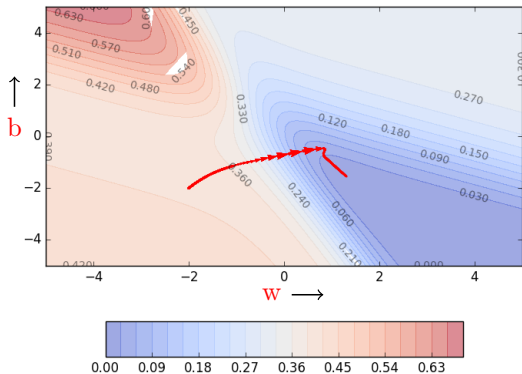
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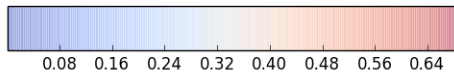
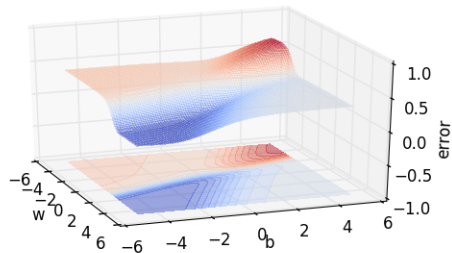


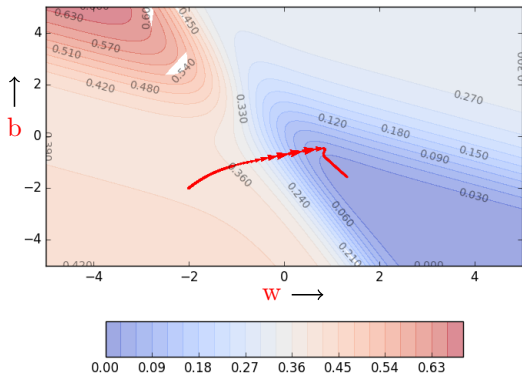
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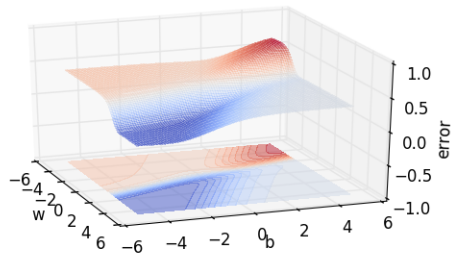


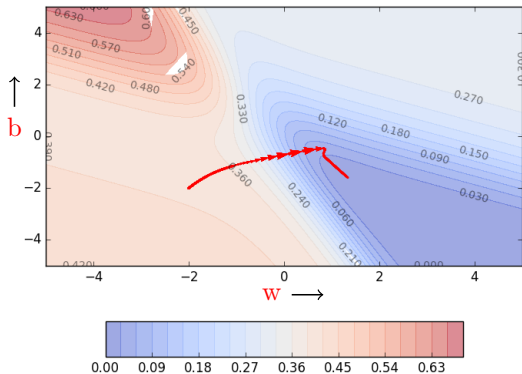
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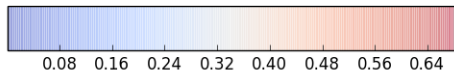
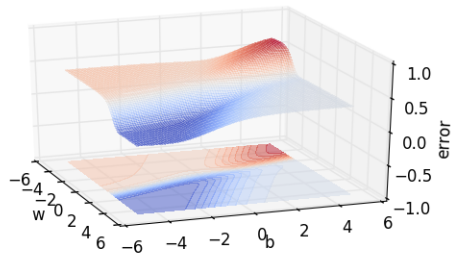


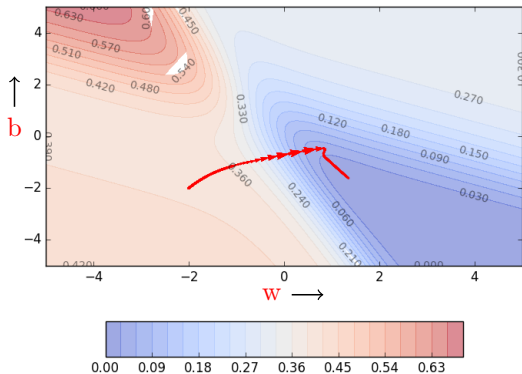
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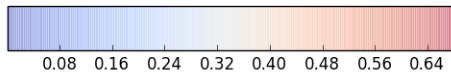
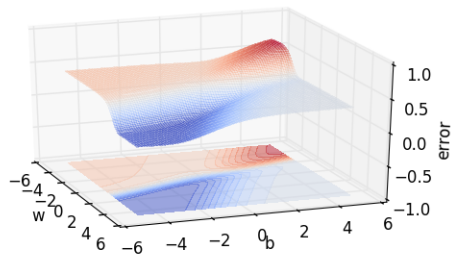


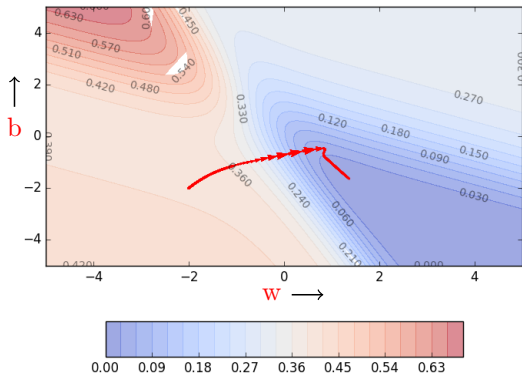
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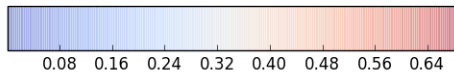
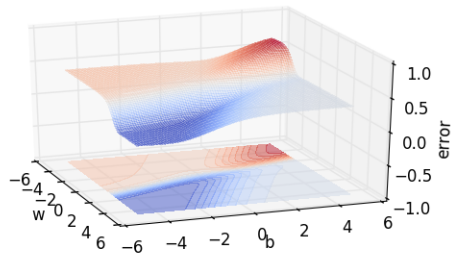


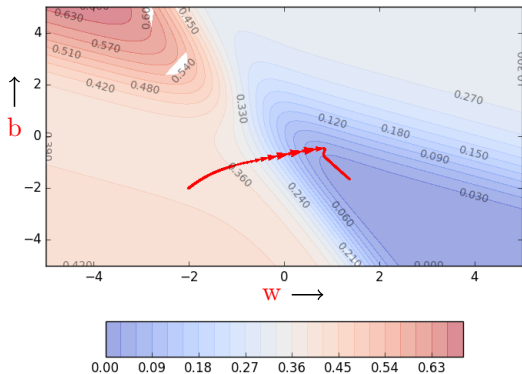
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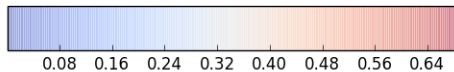
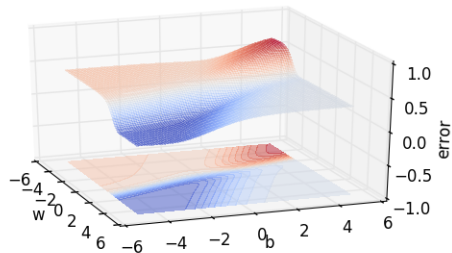


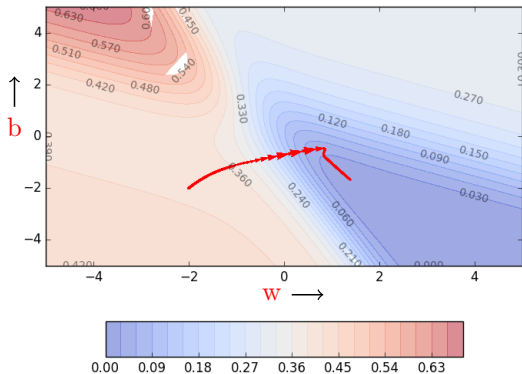
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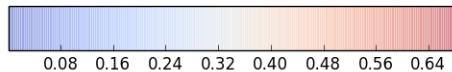
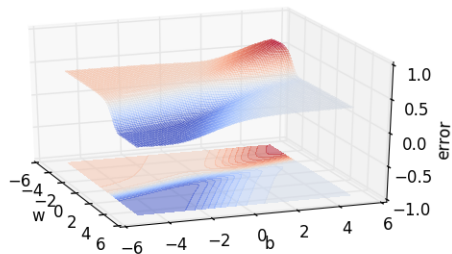


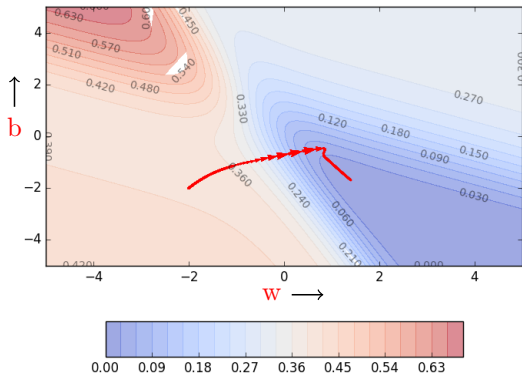
Gradient descent on the error surface



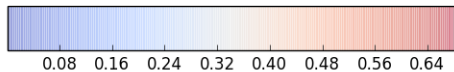
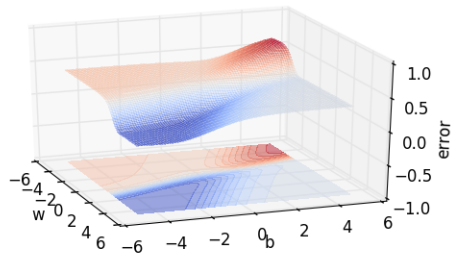


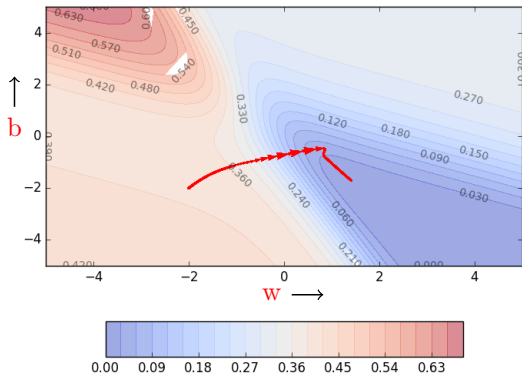
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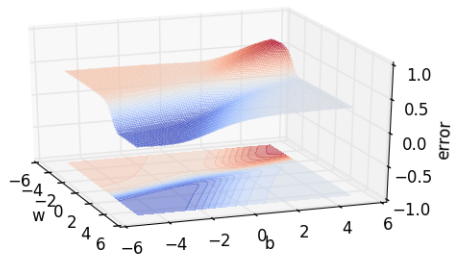


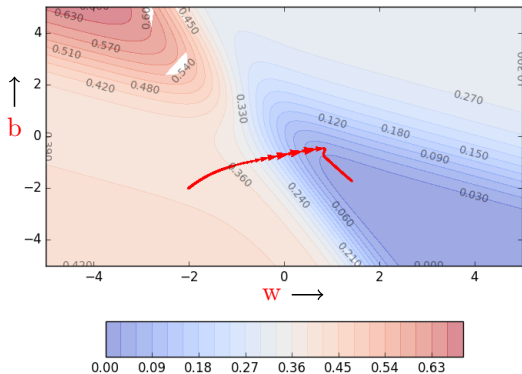
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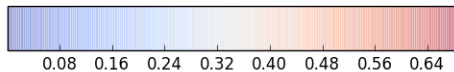
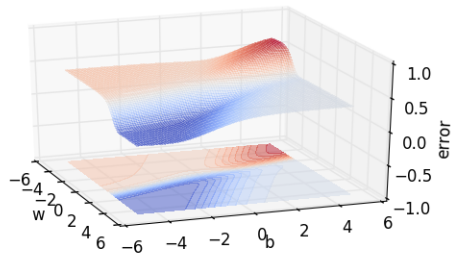


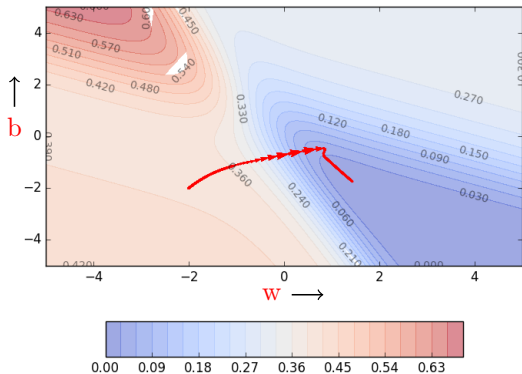
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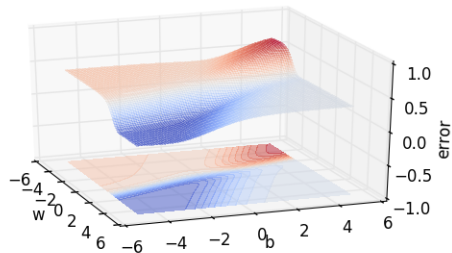


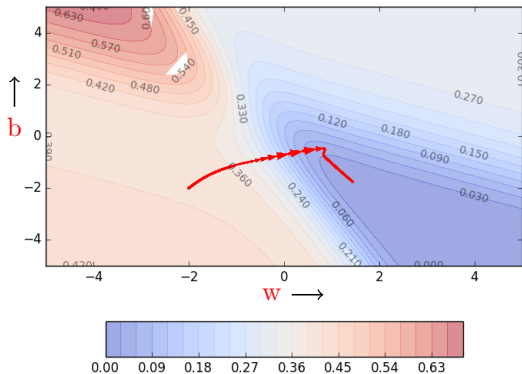
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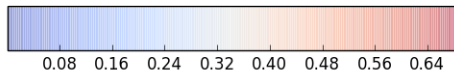
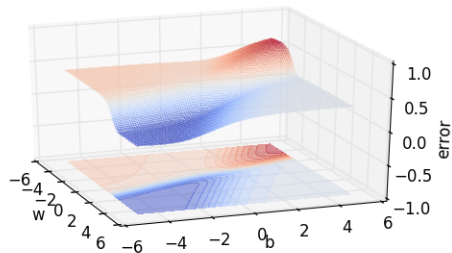


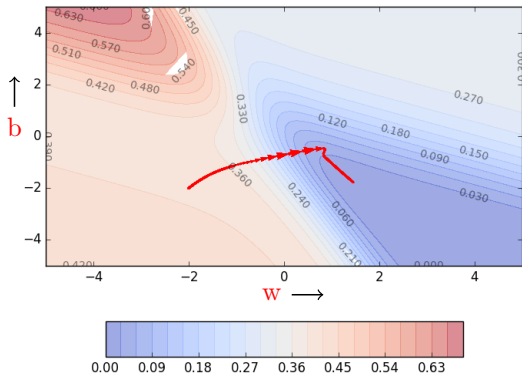
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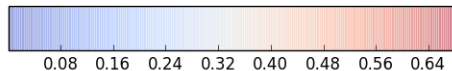
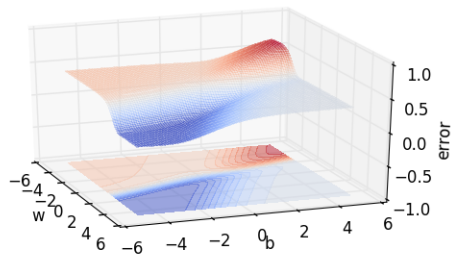


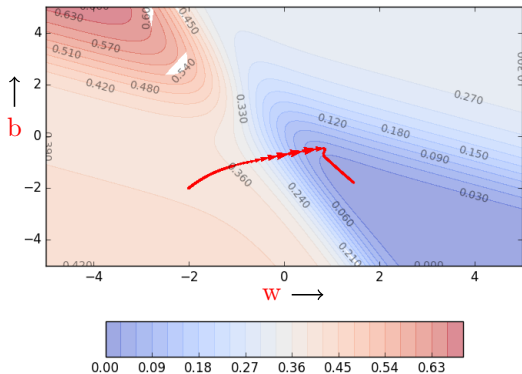
Gradient descent on the error surface



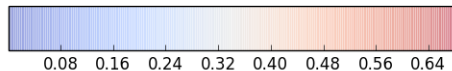
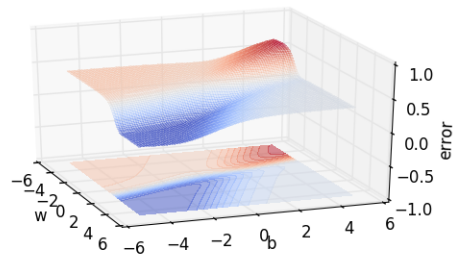


Gradient descent on the error surface





Gradient descent on the error surface



Module 5.4 : Momentum based Gradient Descent

Some observations about gradient descent

- It takes a lot of time to navigate regions having a gentle slope

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- Can we do something better ?

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- It takes a lot of time to navigate regions having a gentle slope
- This is because the gradient in these regions is very small
- Can we do something better ?
- Yes, let's take a look at 'Momentum based gradient descent'

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Update rule for momentum based gradient descent

$$update_t = \gamma \cdot update_{t-1} + \eta \nabla w_t$$

$$w_{t+1} = w_t - update_t$$

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Update rule for momentum based gradient descent

$$update_t = \gamma \cdot update_{t-1} + \eta \nabla w_t$$

$$w_{t+1} = w_t - update_t$$

- In addition to the current update, also look at the history of updates.

$$update_t = \gamma \cdot update_{t-1} + \eta \nabla w_t$$

$$w_{t+1} = w_t - update_t$$

$$update_0 = 0$$

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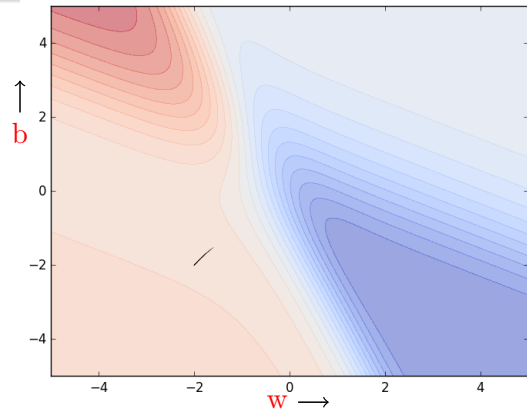
$$update_t = \gamma \cdot update_{t-1} + \eta \nabla w_t = \gamma^{t-1} \cdot \eta \nabla w_1 + \gamma^{t-2} \cdot \eta \nabla w_2 + \dots + \eta \nabla w_t$$

```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
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        v_w = gamma * prev_v_w + eta * dw
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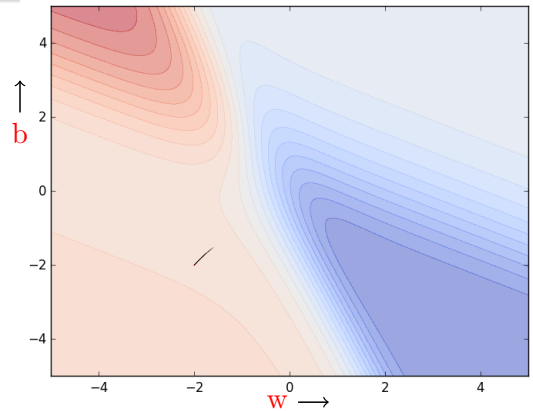


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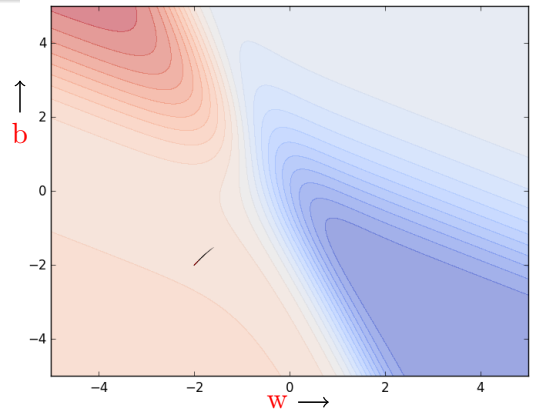


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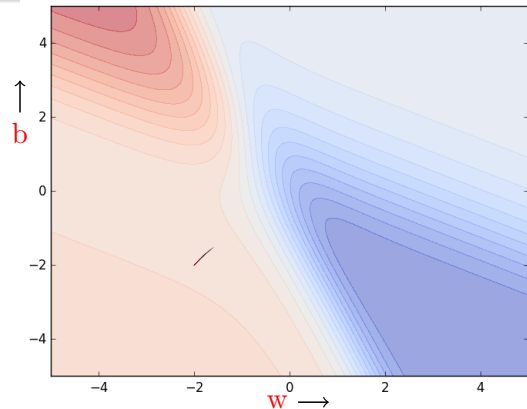


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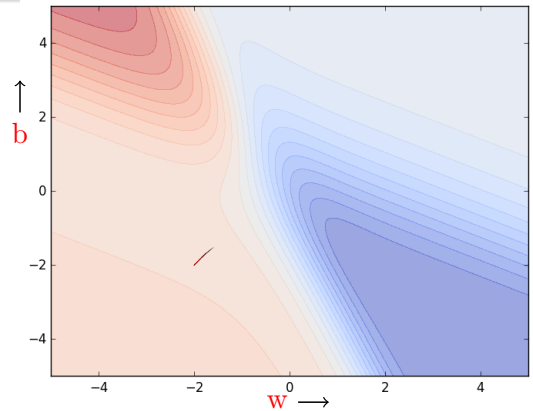


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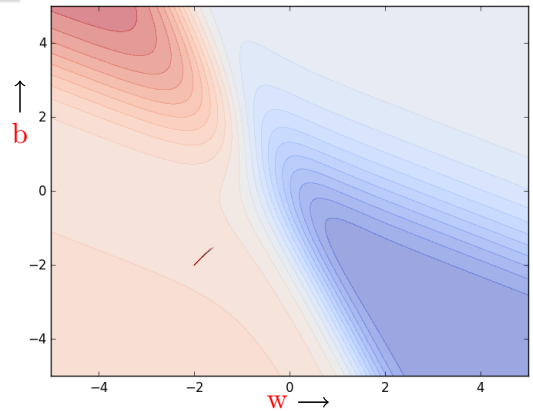


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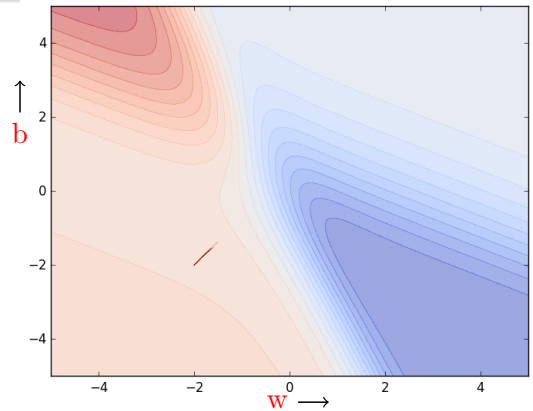


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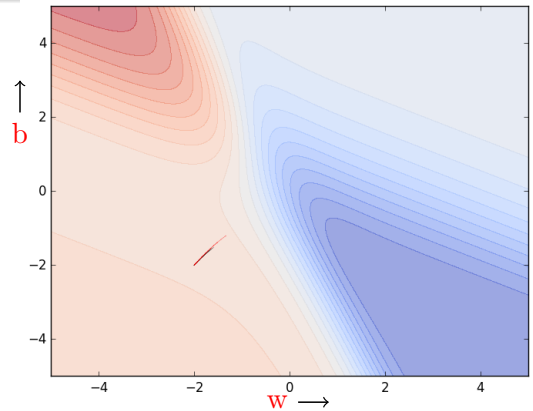


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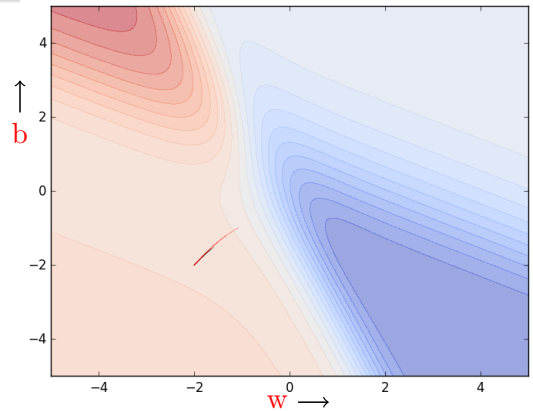


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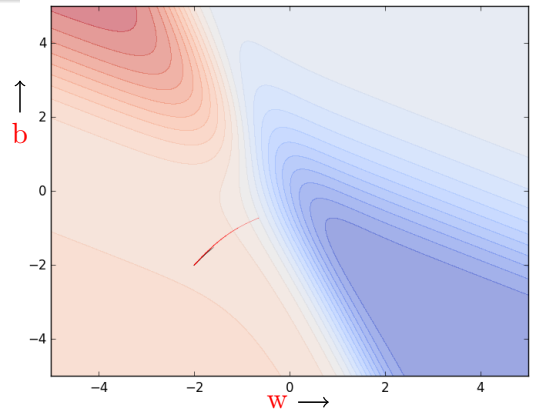


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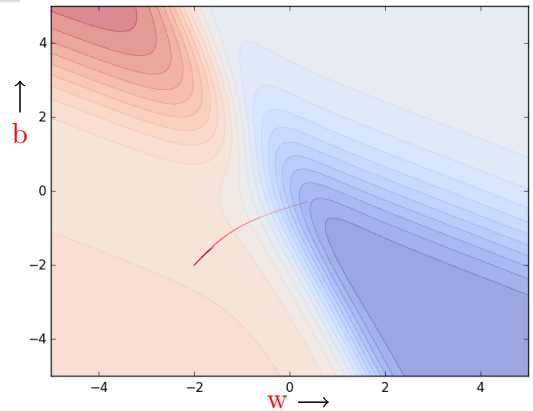


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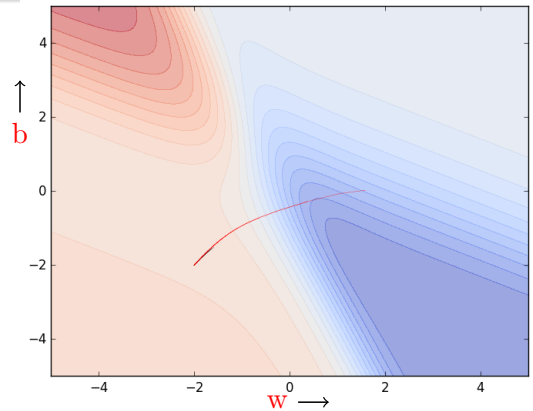


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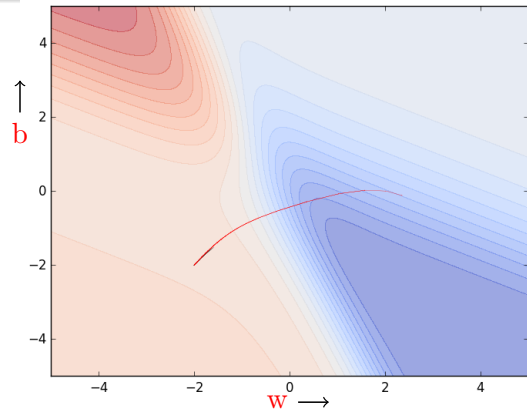


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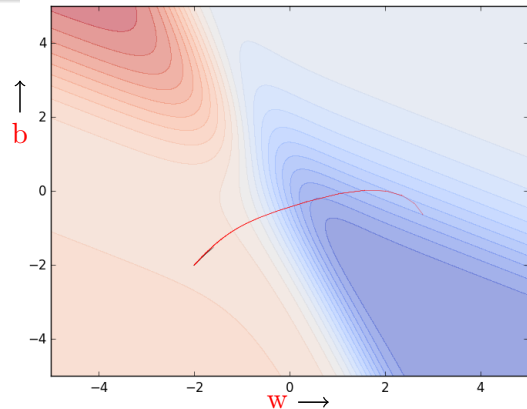


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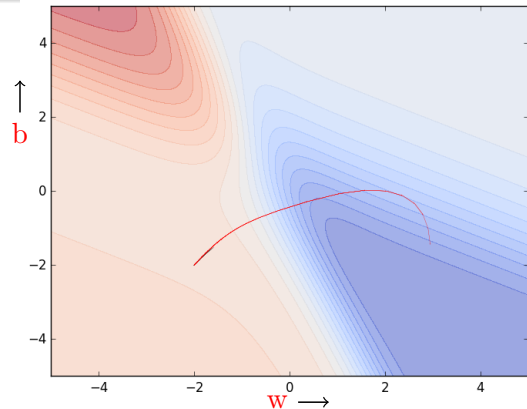


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        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b

```

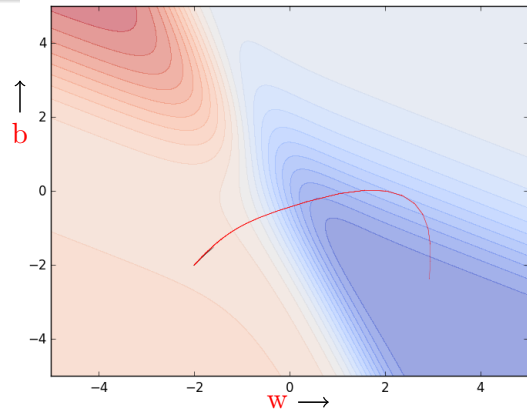


```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b

```

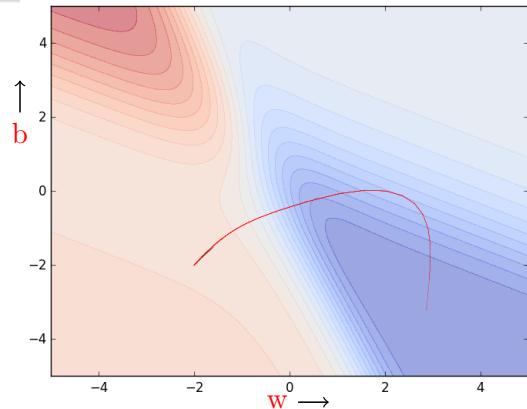


```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
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        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b

```

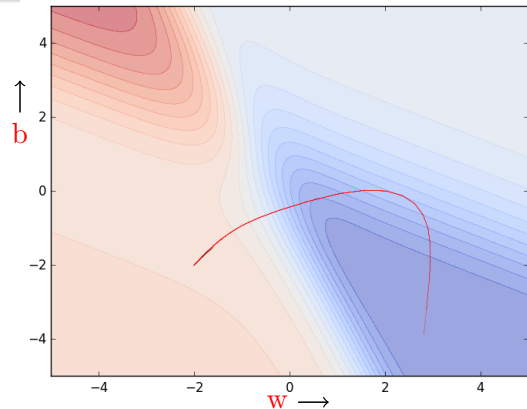


```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
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        prev_v_w = v_w
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```

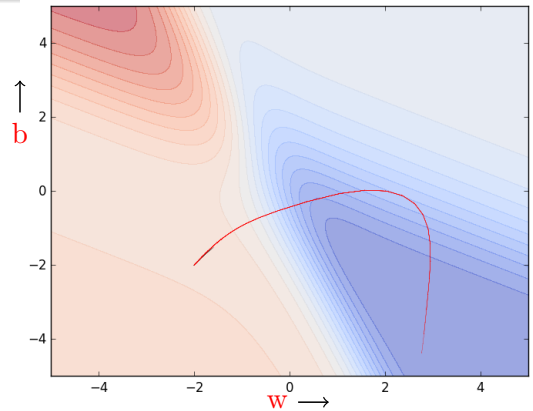


```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
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```

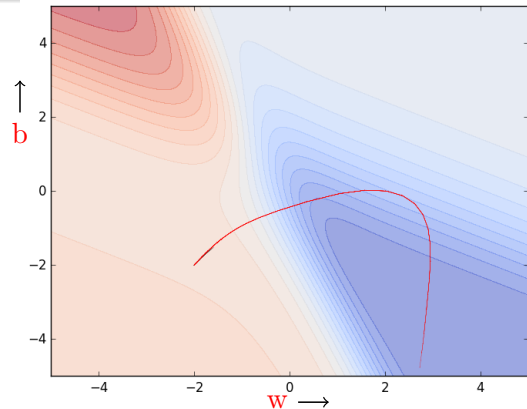


```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
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        w = w - v_w
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        prev_v_w = v_w
        prev_v_b = v_b

```



Some observations and questions

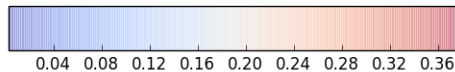
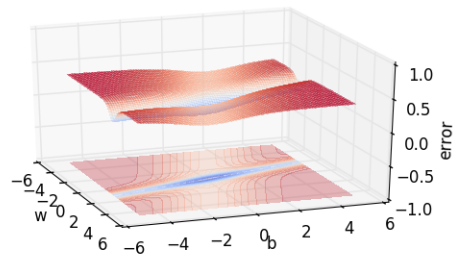
- Even in the regions having gentle slopes, momentum based gradient descent is able to take large steps because the momentum carries it along

Some observations and questions

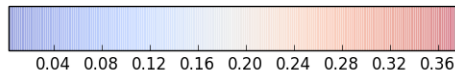
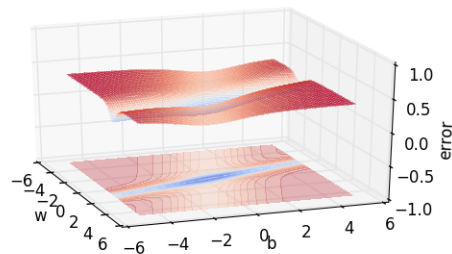
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- Is moving fast always good? Would there be a situation where momentum would cause us to run pass our goal?

Some observations and questions

- Even in the regions having gentle slopes, momentum based gradient descent is able to take large steps because the momentum carries it along
- Is moving fast always good? Would there be a situation where momentum would cause us to run pass our goal?
- Let us change our input data so that we end up with a different error surface and then see what happens ...



- In this case, the error is high on either side of the minima valley



- In this case, the error is high on either side of the minima valley
- Could momentum be detrimental in such cases... let's see....

